Elective - III

UE23AM342BA3 : Explainable AI

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EXPLAINABLE AI

"CAN WE TRUST THE DECISIONS THAT AI MAKES?"

- This is the foundation upon which Explainable AI works.

What is Explainable AI

- With AI systems operating in safety-critical industries, AI prediction has to be justifiable and fair. In the healthcare industry when AI-powered tools detect diseases or recommend treatments, there must be complete trust and understanding of why a model suggested a certain course of action.
- It should justify the treatment plans and validate its decisions with medical evidence. Such criticality can be extrapolated to almost any industry.
- **Explainable Artificial Intelligence (AI) is the ability for artificial intelligence systems to provide understandable explanations for their decisions, recommendations, or predictions.**

Tools/Languages required

- Python Tools and Libraries
- Machine Learning Libraries (Scikit-Learn, TensorFlow/PyTorch)
- XAI Libraries: SHAP, LIME, InterpretML, OmniXAI.

Why Explainable AI Matters in 2025?

•Organizations must be fully aware of the AI decision-making process, and those that establish digital trust among consumers through explainable AI are more likely to see their annual revenue and EBIT grow at rates of 10 percent or more **(Source: McKinsey)**

Here are a few reasons why XAI matters:

- When the stakeholders and regulators understand how the AI arrived at a particular decision, it increases trust and confidence in its recommendation
- XAI can identify and correct the biases related to race, gender, age, and other factors. This results in fair outcomes
- It ensures regulatory compliance as many of them, like the EU's AI Act (GDPR)require explanations for automated decisions that affect people
- It gives context to the outcome it suggests, helping professionals make informed decisions
- When companies understand how a model works, it can help surface business interventions that might have been otherwise hidden

Why Explainable AI is the Future of Responsible Tech?

•AI as a silent decision-maker:
AI increasingly influences critical areas such as loan approvals, medical diagnoses, fraud detection, and autonomous systems, directly impacting human lives.

•Trust is the central challenge:

Explainable AI (XAI) builds transparency:

Regulatory pressure is increasing worldwide:
Laws and frameworks like the **EU AI Act**, **GDPR Article 22**, **NIST AI Risk Management Framework**, and **India's Digital India Act** mandate AI systems to be interpretable, fair, and auditable.

•Non-compliance has serious consequences:
Organizations that fail to meet explainability requirements risk heavy fines, legal action, and regulatory bans.

•Growing demand for XAI expertise:

Compliance – a company adhering to applicable rules and laws

High-stakes domains cannot rely on black-box AI:

•Explainability is essential for AI adoption in healthcare:
Clinicians must understand and justify AI-assisted diagnoses; lack of explainability remains a major barrier to scaling AI in healthcare (McKinsey).

•XAI enables cross-functional collaboration:
Explainability aligns data scientists, product managers, and compliance teams around a shared understanding of how AI systems behave.

•A shared source of truth for responsible AI:
XAI bridges silos, ensuring AI systems meet both performance objectives and ethical, legal, and regulatory requirements.

Emerging Roles in XAI Careers:

- **Explainable AI Engineer:**
- **AI Ethics Specialist:**
- **Data Scientist (XAI Focus):**
- **Model Governance Analyst:**
- **AI Policy and Compliance Consultant:**
- **XAI Research Scientist:**

How Explainable AI is Used Across Industries: HFRCC

Healthcare:

The AI models that are used in the healthcare industry must provide clear explanations for their recommendations. The physicians must be convinced that the basis on which the AI made the decision is solid and there is no room for error. For example, the AI system can share the specific medical evidence and patient data that it used to suggest treatment to a patient. The doctor can validate the AI's reasoning and take it forward.

Finance:

Explainable AI can detect fraudulent activities by explaining how certain transactions are flagged as suspicious. When there is a clear criteria being used for its reasoning, it will ensure fairness and reduce biases. For example: It can provide banks with clear examples on why someone's loan application was rejected.

Recruitment:

There are many companies that use AI systems to screen a large volume of job applications. XAI tools can help reveal the biases (if any) that are in the AI-driven algorithms used in hiring.

Computer Vision:

Explainable AI techniques can be used to gather insights into factors that are relevant and influential in recognizing and classifying images.

Cybersecurity:

The AI algorithms that are used to detect threats must give the rationale behind each alert. With Explainable AI, professionals will be able to understand and trust the reasoning behind the alerts.

Challenges in Explainable AI:

Complexity of the AI:

One of the main challenges is the complexity of the AI itself. The AI that we talk about generally means machine learning. They become smarter when more and more data is fed into it. It requires complex mathematical models that can be difficult to translate into explanations that the average layman might not understand.

Explainability Requires a Performance Tradeoff:

Most machine algorithms are coded in a manner where it provides the results as fast and efficiently as possible. It doesn't use its resources to explain how it arrived at a particular recommendation. They achieve high accuracy by learning intricate patterns, whose complexity sacrifices transparency.

No Standard XAI Evaluation Methods:

The explanations that you receive from XAI depend on the techniques that are used. This makes it difficult to validate the correctness. Also, evaluations are domain-specific. The explanations that are suitable for the insurance industry might not meet the criteria for medical cases.

Addressing such challenges will need a consistent evaluation framework and require tailoring of expectations to specific user needs.

Future Trends in XAI:

Rise of Responsible AI Ops:

Responsible AI Ops blends AI ML Ops with frameworks for transparency and ethical risk monitoring across the entire AI lifecycle. It embeds explainability into deployment pipelines. It keeps the AI continuously accountable and adaptable to regulatory changes.

LLMs with Explainability Features:

LLMs are at the core of generative AI and automation, but the lack of transparency is concerning. Researchers are working on enhancing these models with built-in explainability modules. It provides the users with the rationale for which data influenced its answers and it even allows the users to interrogate the reasoning steps.

Explainability in Generative AI:

Generative AI is evolving into multimodal systems where it can handle all the data types together. Enterprises that deploy generative AI for design, drug discovery, or autonomous systems are working on developing layered explainability tools. The tools will show why a generative model created a particular output across modalities. This interpretability is crucial for business accountability.

Human-in-the-Loop AI Governance Models:

With Explainable AI on the rise, it is evolving into human-in-the-loop where the AI systems work with human experts. This shift is mainly because of regulatory mandates, especially in healthcare and finance.

Explainable AI is a significant component toward responsible technology.

Industry 4.0 and 5.0

XAI isn't just a feature; it's the fundamental enabler for moving from efficient, but impersonal, Industry 4.0 to a more responsible, creative, and human-focused Industry 5.0

Explainable AI-Evaluation Policy

Category	Activity	Marks	Marks Reduced To
Theory	**ISA 1**	40	15
Theory	**ISA 2**	40	15
Experiential Learning Component - Banana problem	ISA - Implementation Based (conducted in the class)	4	4
Experiential Learning Component - Orange problem	ISA - Implementation Based (conducted in the class)	6	6
Experiential Learning – Jackfruit	ISA - Course Project	10	10
Theory	**ESA**	100	50

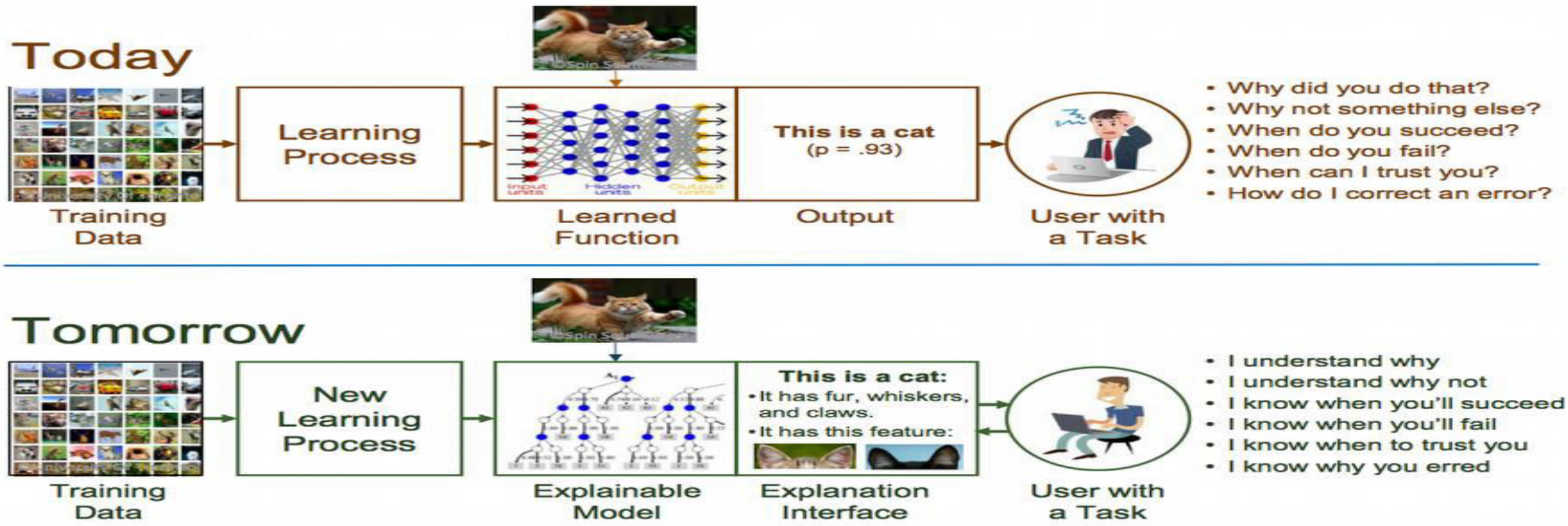
UNIT 1: Introduction to Interpretability and Interpretable Models

MOTIVATION

- **What is Explainable AI ?**

Explainable Artificial Intelligence (AI) is the ability for artificial intelligence

systems to provide understandable explanations for their decisions, recommendations, or predictions



Interpretability Vs Explainability

Term	Definition	Source
Interpretability	*"level of understanding how the underlying (AI) technology works"*	ISO/IEC TR 29119-11:2020(en), 3.1.42[35]
Explainability	*"level of understanding how the AI-based system ... came up with a given result"*	ISO/IEC TR 29119-11:2020(en), 3.1.31[35]

Source: https://en.wikipedia.org/wiki/Explainable_artificial_intelligence

- **AI Interpretability** Focuses on understanding the innerworking of an AI Model (About Transparency of AI model)
- **AI Explainability** aims to provide reasons for models output(Justification of Model's Output)

Interpretability Vs Explainability

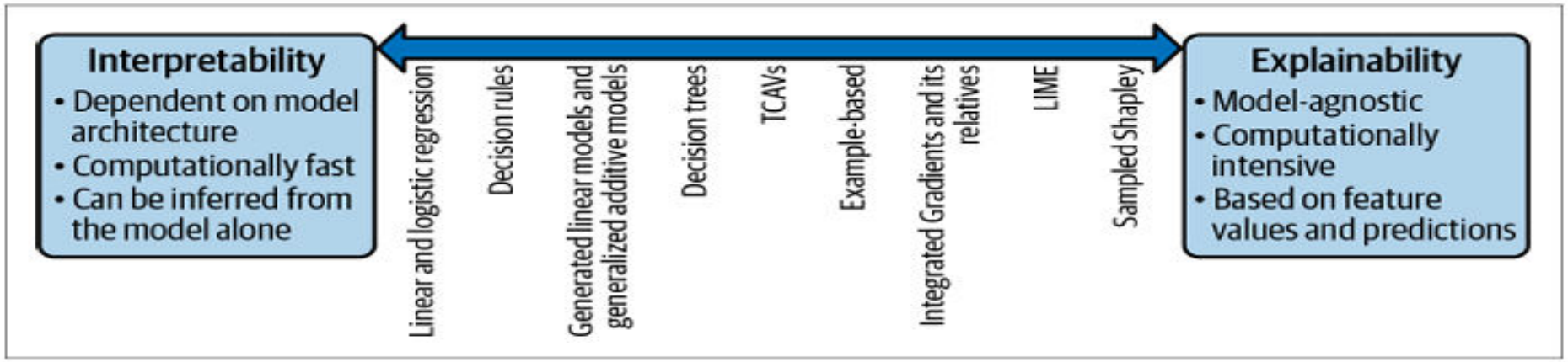


Figure 2-1. Explainability and interpretability are two ends of a spectrum. Here we show key characteristics of techniques at each end of the spectrum, and show some examples of where techniques fall along this spectrum.

- **AI Interpretability** Focuses on understanding the innerworking of an AI Model (About Transparency of AI model)
- **AI Explainability** aims to provide reasons for models output(Justification of Model's Output)

Interpretability Vs Explainability

While often used interchangeably, interpretability and explainability represent fundamentally different approaches to understanding model behavior. This distinction shapes both model selection and the tools we use to audit AI systems.

Interpretability

Direct understanding of model mechanics - the ability to comprehend how a model works by examining its structure and parameters directly. Linear regression and decision trees are inherently interpretable because humans can trace the path from input to output.

Key characteristic: Transparency is built into the model architecture itself, requiring no additional tools or post-processing.

Explainability

Post-hoc justification of predictions - techniques applied after training to understand what a black-box model has learned. Methods like SHAP and LIME approximate model behavior without revealing internal mechanisms.

Key characteristic: External tools that provide insights into opaque models, though these explanations may be approximations.

The trade-off is clear: interpretable models offer genuine transparency but may sacrifice predictive power, while explainability tools enable complex models at the cost of indirect understanding. Your choice depends on domain requirements, regulatory constraints, and acceptable risk levels.

The Interpretability Spectrum: From Transparent to Opaque

Machine learning models exist on a continuum from fully transparent to completely opaque. Understanding where your model falls on this spectrum is crucial for selecting appropriate explanation methods and setting stakeholder expectations.

Decision Trees

Completely transparent and interpretable. Every decision path is explicit and traceable. Stakeholders can follow the exact logic: "If age > 35 AND income > $50k, then approve."

Deep Neural Networks

Black-box models that require external explanation layers. Despite high accuracy, the internal transformations across hidden layers remain opaque without tools like DeepLIFT, GradCAM, or attention visualization.

The DeepLIFT Framework: Deep Learning Important FeaTures decomposes neural network predictions by comparing activations to reference values, revealing which input features most influenced the output. Unlike gradient-based methods, DeepLIFT handles activation saturation more effectively.

The interpretability gap between these model types isn't just technical—it has real-world implications for debugging, compliance, and user trust. Choose wisely based on your domain's tolerance for opacity.

Case Study: Loan Approval Decision Systems

Consider a financial institution deciding whether to approve a $50,000 personal loan. The choice between interpretable and black-box models has profound implications for both regulatory compliance and customer trust.

Decision Tree Approach

Transparent rules for approval

- IF credit_score ≥ 720
- AND debt_to_income < 0.43
- AND employment_years ≥ 2
- THEN approve_loan

The applicant and loan officer can see exactly why the decision was made. If denied, the customer knows precisely which criteria to improve. This transparency builds trust and satisfies regulatory requirements automatically.

Black-Box Model Approach

Requires SHAP or LIME for explanation

A gradient boosting model or neural network might achieve 3-5% higher accuracy but offers no inherent transparency. Post-hoc explanation reveals:

- Credit score: +0.32 impact
- Recent credit inquiries: -0.18 impact
- Account age: +0.15 impact

These approximations provide insight but may not satisfy regulatory scrutiny or fully explain model behavior on edge cases.

The decision tree sacrifices 3% accuracy for complete transparency—a worthwhile trade in regulated industries. The black-box model gains predictive power but introduces explanation complexity and potential compliance risks.

KEY TAKEAWAY

Strategic Guidelines for Model Selection

Prioritize Interpretable Models When Possible

In high-stakes domains like healthcare, finance, and criminal justice, start with inherently interpretable models. Linear models, decision trees, and rule-based systems provide transparency that black-box models cannot match, even with sophisticated explanation tools.

Best for: Regulated industries, critical decisions, domains requiring audit trails, situations where accuracy differences are minimal

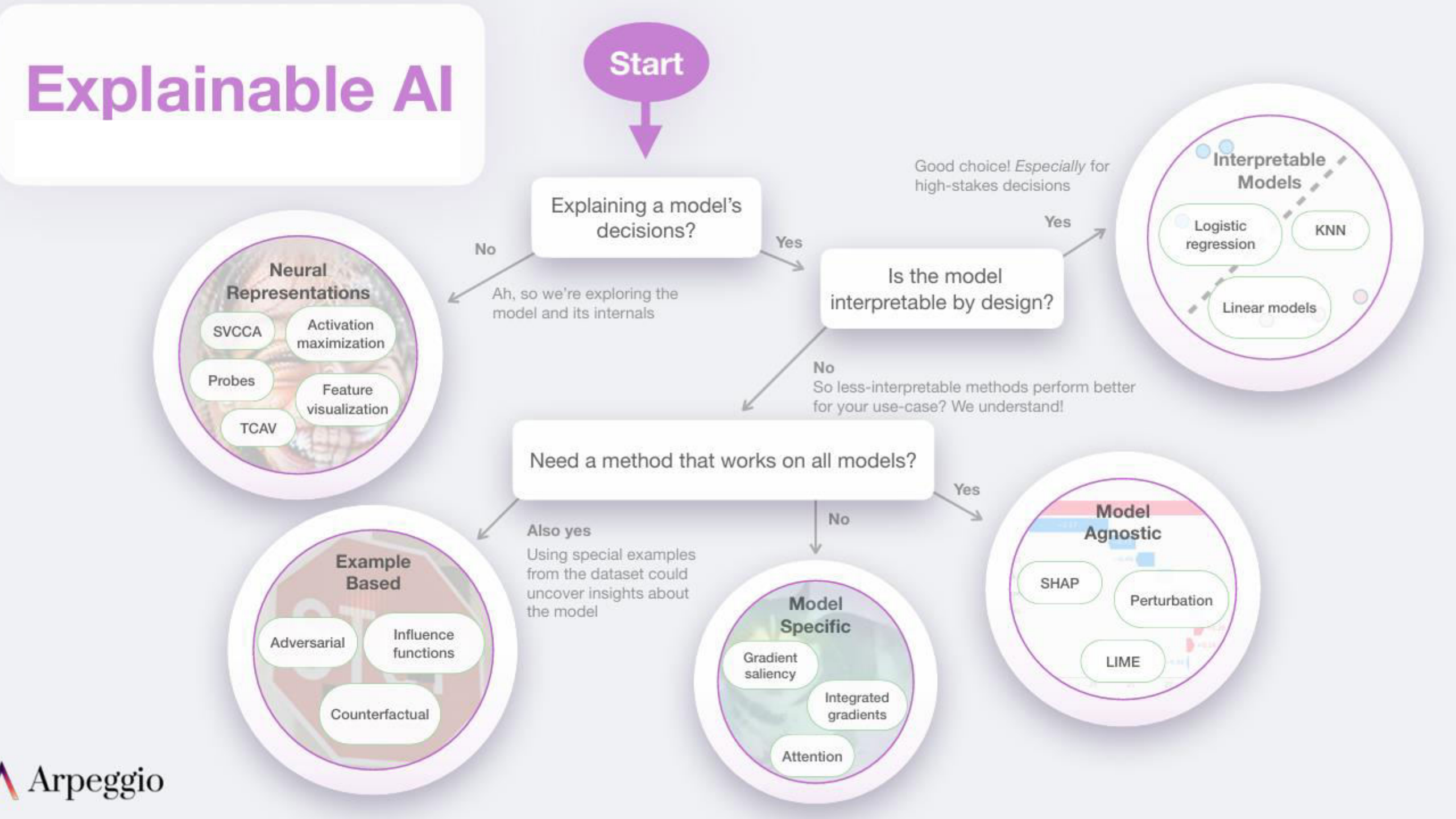
Deploy Explainability Tools for Complex Models

When interpretable models lack sufficient predictive power, leverage state-of-the-art explainability frameworks. SHAP provides theoretically grounded feature importance, LIME offers local approximations, and attention mechanisms reveal what neural networks focus on.

Best for: Computer vision, natural language processing, scenarios where accuracy gains justify complexity, research settings

"The best model is not always the most accurate one—it's the one that stakeholders can trust, regulators can audit, and practitioners can debug"

Explainable AI



Arpeggio

Categorization in XAI

Model-**agnostic**	Applicable to all model types
Model-**specific**	Only applicable to a specific model type

Agnosticity

Global explanation	Explaining the whole model
Local explanation	Explaining individual predictions

Scope

Graph	Image
Text / Speech	Tabular

Data type

Visual	Feature importance
Data points	Surrogate models

Explaination type

Taxonomy of Interpretability Methods

Understanding the landscape of interpretability approaches requires organizing methods along multiple dimensions. This taxonomy helps practitioners select the right technique for their specific use case and constraints.

Intrinsic vs Post-hoc

Does interpretability come from the model architecture itself, or is it added after training through external analysis tools?

Global vs Local

Are we explaining the model's overall behavior across all inputs, or understanding a single specific prediction?

Transparent vs Black-box

Can humans directly understand the model's decision process, or do we need approximation methods to peek inside?

These dimensions are not mutually exclusive—a decision tree is both intrinsic and transparent, while SHAP provides post-hoc explanations that can be either global or local. Understanding these categories empowers you to match explanation methods to stakeholder needs and technical constraints.

Intrinsic vs Post-hoc Interpretability

The fundamental divide in interpretability approaches centers on *when* and *how* we achieve understanding of model behav
This distinction affects everything from model selection to audit procedures.

Intrinsic : Interpretable by design

The model structure itself is transparent and human-comprehensible. Examples include linear regression, decision trees, rule-based systems, and sparse additive models.

- No additional tools needed
- Explanation is exact, not approximate
- Often preferred for regulated domains

1

2

Post-Hoc: Explanation generated after prediction

External methods analyze trained models to approximate their behavior. Techniques include SHAP, LIME, attention visualization, and saliency maps.

- Enables complex model architectures
- Explanations may be approximations
- Requires additional computational overhead

Trade-offs to Consider

Intrinsic models guarantee fidelity—the explanation is the model. However, they may sacrifice predictive accuracy for transparency.

Post-hoc methods unlock powerful architectures but introduce approximation error. The explanation might not fully capture model complexity or edge case behavior.

Global vs Local Explanations: Scope Matters

Beyond the intrinsic versus post-hoc distinction, interpretability methods differ in their scope of analysis. Understanding whether you need global or local explanations shapes your choice of tools and techniques.

Global Explanations

Understanding entire model behavior

Global methods reveal how the model behaves across the entire feature space and dataset. They answer questions like "What features matter most overall?" and "What patterns has the model learned?"

Examples: Feature importance rankings, partial dependence plots, global SHAP values, model-wide rule extraction

Use cases: Model debugging, bias auditing, regulatory compliance, stakeholder communication, scientific discovery

Local Explanations

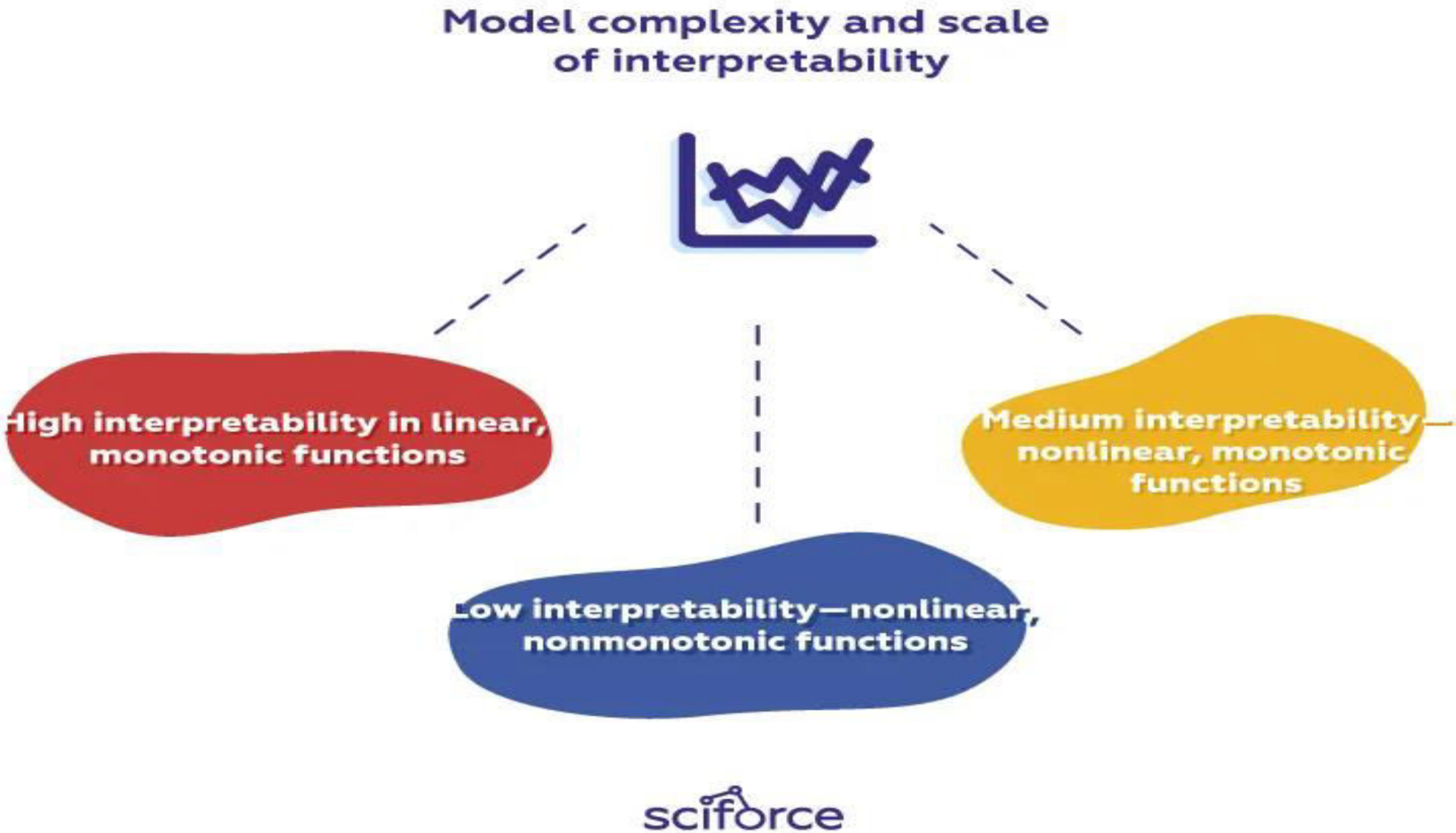
Understanding individual predictions

Local methods explain why the model made a specific prediction for a particular instance. They answer "Why was this loan denied?" or "What made this transaction flagged as fraudulent?"

Examples: LIME, individual SHAP values, counterfactual explanations, attention weights, influence functions

Use cases: Customer explanations, appeal processes, debugging specific failures, personalized recommendations

Complementary Approaches: The most robust interpretability strategies combine both global and local methods. Global analysis identifies systemic issues and overall model behavior, while local explanations help users understand specific decisions that affect them directly.


Model complexity and scale of interpretability
High interpretability in linear, monotonic functions
Medium interpretability— nonlinear, monotonic functions
Low interpretability—nonlinear, nonmonotonic functions
sciforce

Case Study: Decision Trees vs SHAP Explanations in Practice

Let's examine a real-world scenario comparing intrinsic interpretability against post-hoc explainability, highlighting the practical trade-offs that data scientists face when deploying models to production.

Decision Tree Model

Performance: 82% accuracy

Interpretability: Complete transparency

- 15 decision rules easily understood by stakeholders
- Zero explanation overhead
- Instant regulatory compliance
- Customers can contest decisions with clear criteria
- Model debugging takes minutes

Limitation: Cannot capture complex non-linear interactions that improve accuracy

Gradient Boosting + SHAP

Performance: 89% accuracy

Interpretability: Post-hoc approximation

- 7% accuracy improvement
- SHAP values require 2-5 seconds per prediction
- Explanations may vary with background dataset
- Harder to validate explanation fidelity
- Additional complexity in deployment

Advantage: Captures subtle patterns that simpler models miss

7%	100%	5s
Accuracy Gain	Transparency	Explanation Time
Performance improvement from complex model	Decision tree offers complete interpretability	Additional latency for SHAP computation

> **The Verdict:** In regulated financial services, the decision tree's transparency often outweighs the accuracy gain. For recommendation systems or fraud detection where false negatives are costly, the complex model with explanations may be justified. Context determines the right choice.

Evaluations of Interpretability

Evaluating machine learning interpretability involves assessing how well humans can understand a model's decisions, often through **application-grounded** (real-world tasks), **human-grounded** (user studies comparing explanations), or **functionally-grounded** (proxy metrics like rule count) methods

Application-grounded evaluation
(real task)
Requires a human to perform experiments within a real-life application.
For example, to evaluate an interpretation on diagnosing a certain disease, the best way is for the doctor to perform diagnoses.

Human-grounded evaluation
(simple task)
is about conducting simpler human-subject experiments.
For example, humans are presented with pairs of explanations, and must choose the one that they find to be of higher quality.

Functionally grounded evaluation
Requires no human experiments.
Instead, it uses a proxy for explanation quality.
This method is much less costly than the previous two. The challenge, of course, is to determine what proxies to use.
For example, decision trees have been considered interpretable in many situations, but additional research is required.

Scope of Interpretability

The scope of interpretability in artificial intelligence and machine learning defines the extent and level at which a human can understand a model's inner workings, its decision-making process, or the rationale behind a specific prediction. It ranges from understanding the entire system's logic to explaining a single outcome and is vital for building trust, ensuring fairness, meeting regulations, and debugging models.

- **Algorithm Transparency**
  How does the algorithm create the model?
- **Global, Holistic Model Interpretability**
  How does the trained model make predictions?
- **Global Model Interpretability on a Modular Level**
  How do parts of the model affect predictions?
- **Local Interpretability for a Single Prediction**

Why did the model make a certain prediction for an instance?

- **Local Interpretability for a Group of Predictions** Why did the model make specific predictions for a group of instances?

THANK YOU

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INTERPRETABLE MODELS

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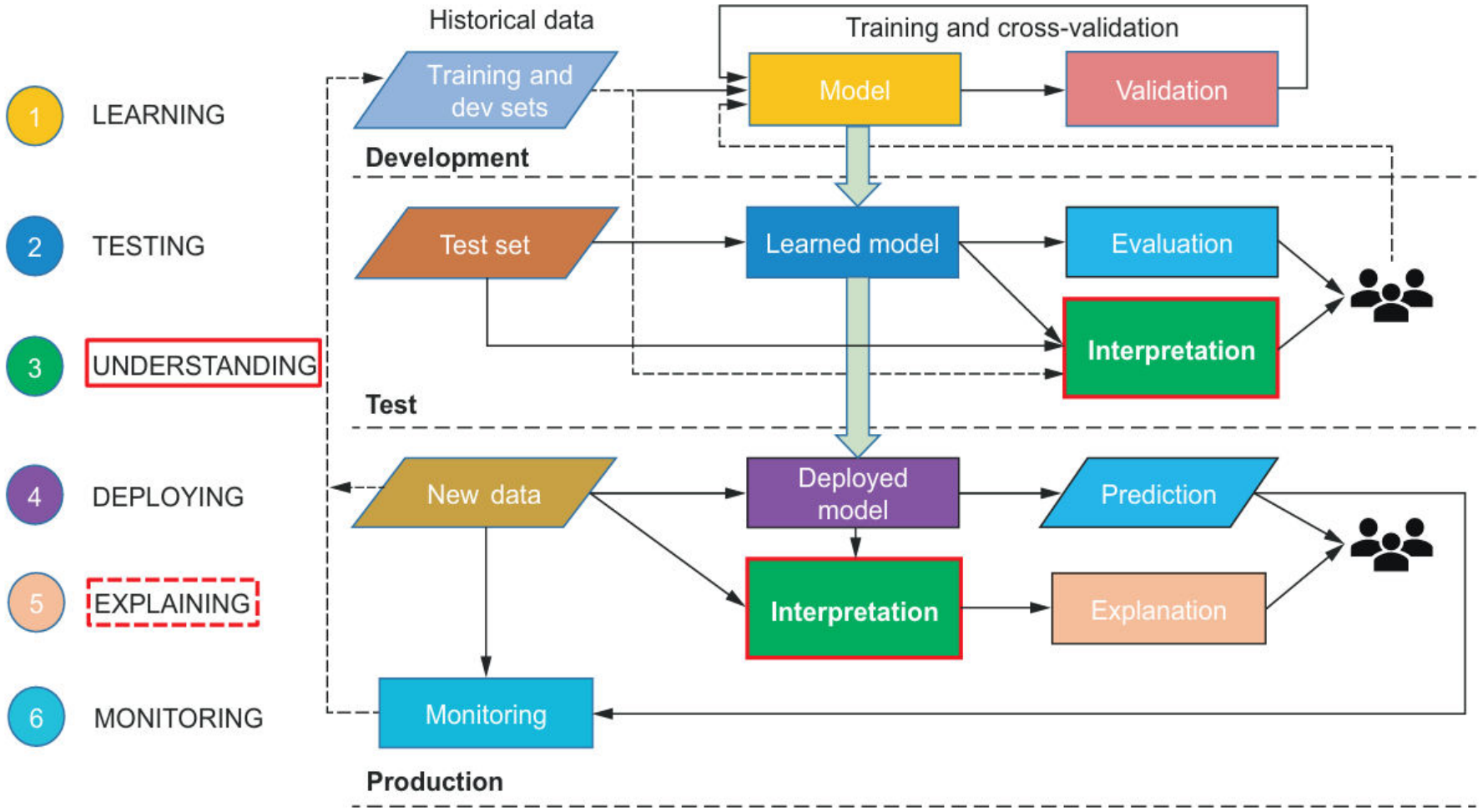
Interpretable Models:

WHITE-BOX MODELS : INHERENTLY TRANSPARENT AND INTERPRETABLE

How to build a robust, interpretable AI system.?



Whitebox models:

White-box models are inherently transparent, and the characteristics that make them transparent are:

- The algorithm used for machine learning is straightforward to understand, and we can clearly interpret how the input features are transformed into the output or target variable.
- We can identify the most important features to predict the target variable, and those features are understandable
- **Eg: Linear regression, Logistic Regression,Decision Trees, GAM(generalized additive models), GLM(Generalized Linear Models)**

White-box model	Machine learning task(s)
Linear regression	Regression
Logistic regression	Classification
Decision trees	Regression and classification
GAMs	Regression and classification

Linear Regression:

- A linear regression model predicts the target as a weighted sum of the feature inputs.

  The linearity of the learned relationship makes the interpretation easy. Linear models can be used to model the dependence of a regression target y on some features x.

- The learned relationships are linear and can be written for a single instance i as follows:

$$y = w_0 + w_1 x_1 + w_2 x_2 + \ldots + w_n x_n$$
$$= w_0 + \sum_{i=1}^{n} w_i x_i$$

The objective of the linear regression learning algorithm is to determine the weights that accurately predict the target variable for all patients in the training set. We can apply the following techniques here:

- **Gradient descent**
- **Closed-form solution (e.g., the Newton equation)**

Linear Regression : Example

•A linear regression model can be easily trained using the Scikit-Learn package in Python. The code to train the model is shown next. Note that the open diabetes dataset provided by Scikit-Learn is used here

Imports numpy to evaluate the performance of model

Imports the scikit-learn class for linear regression

Imports the scikit-learn function to split the data into training and test sets

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
import numpy as np

X_train, X_test, y_train, y_test = train_test_split(X, y,
  test_size=0.2,
  random_state=42)

lr_model = LinearRegression()

lr_model.fit(X_train, y_train)

y_pred = lr_model.predict(X_test)

mae = np.mean(np.abs(y_test - y_pred))
```

Splits the data into training and test sets, where 80% of the data is used for training and 20% of the data for testing, and ensures that the seed for the random-number generator is set using the random_state parameter to ensure consistent train-test splits

Initializes the linear regression model, which is based on least squares

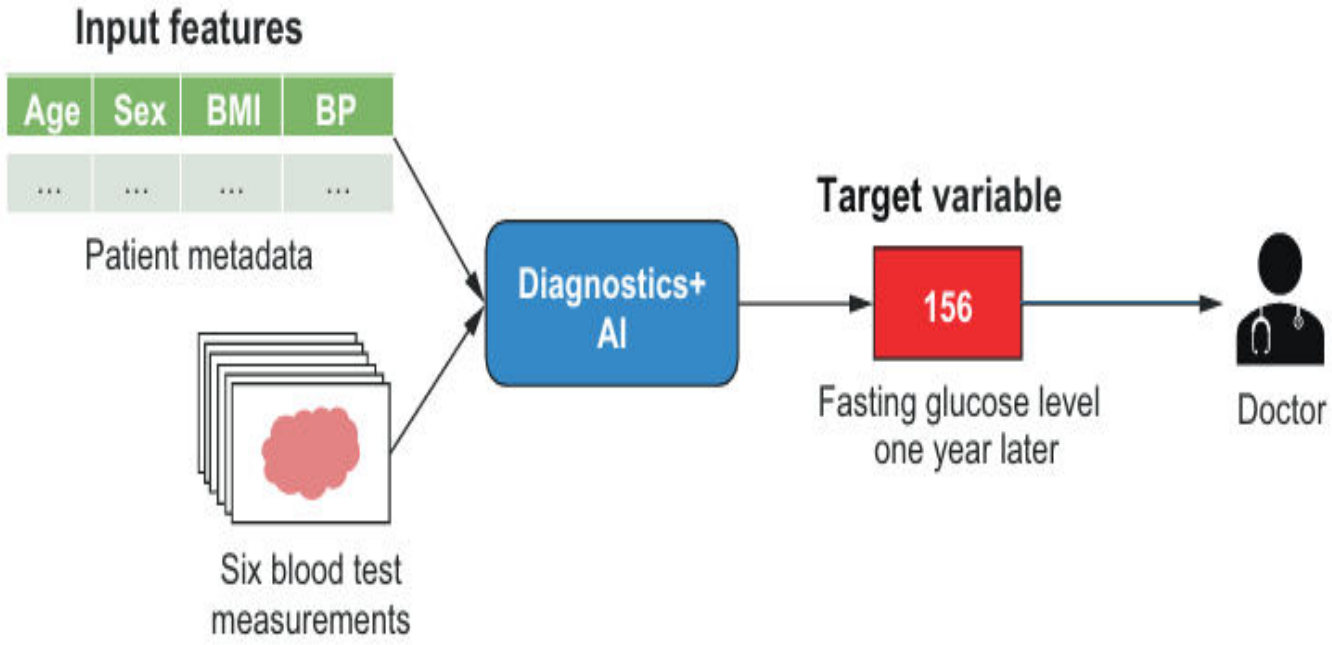
Learns the weights for the model by fitting on the training set

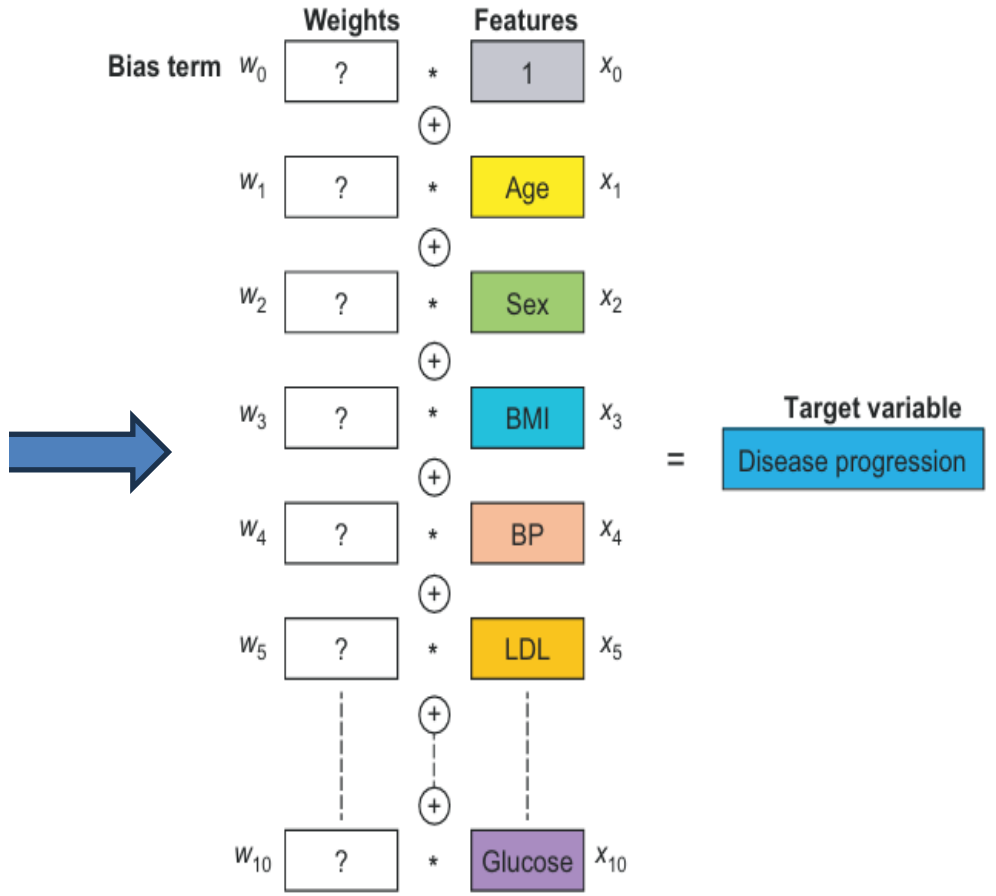
Uses the learned weights to predict the disease progression for patients in the test set

Evaluates the model performance using the mean absolute error (MAE) metric

Interpreting linear regression : Case study

Diabetes Prediction Example:
Disease Progression represented as linear comination of 10 features





Interpreting linear regression : Case study

Diabetes Prediction Example:

```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
sns.set(style='whitegrid')
sns.set_palette('bright')

weights = lr_model.coef_

feature_importance_idx = np.argsort(np.abs(weights))[::-1]
feature_importance = [feature_names[idx].upper() for idx in
  feature_importance_idx]
feature_importance_values = [weights[idx] for idx in
  feature_importance_idx]

f, ax = plt.subplots(figsize=(10, 8))
sns.barplot(x=feature_importance_values, y=feature_importance, ax=ax)
ax.grid(True)
ax.set_xlabel('Feature Weights')
ax.set_ylabel('Features')
```

Imports numpy to perform operation on vectors in an optimized way

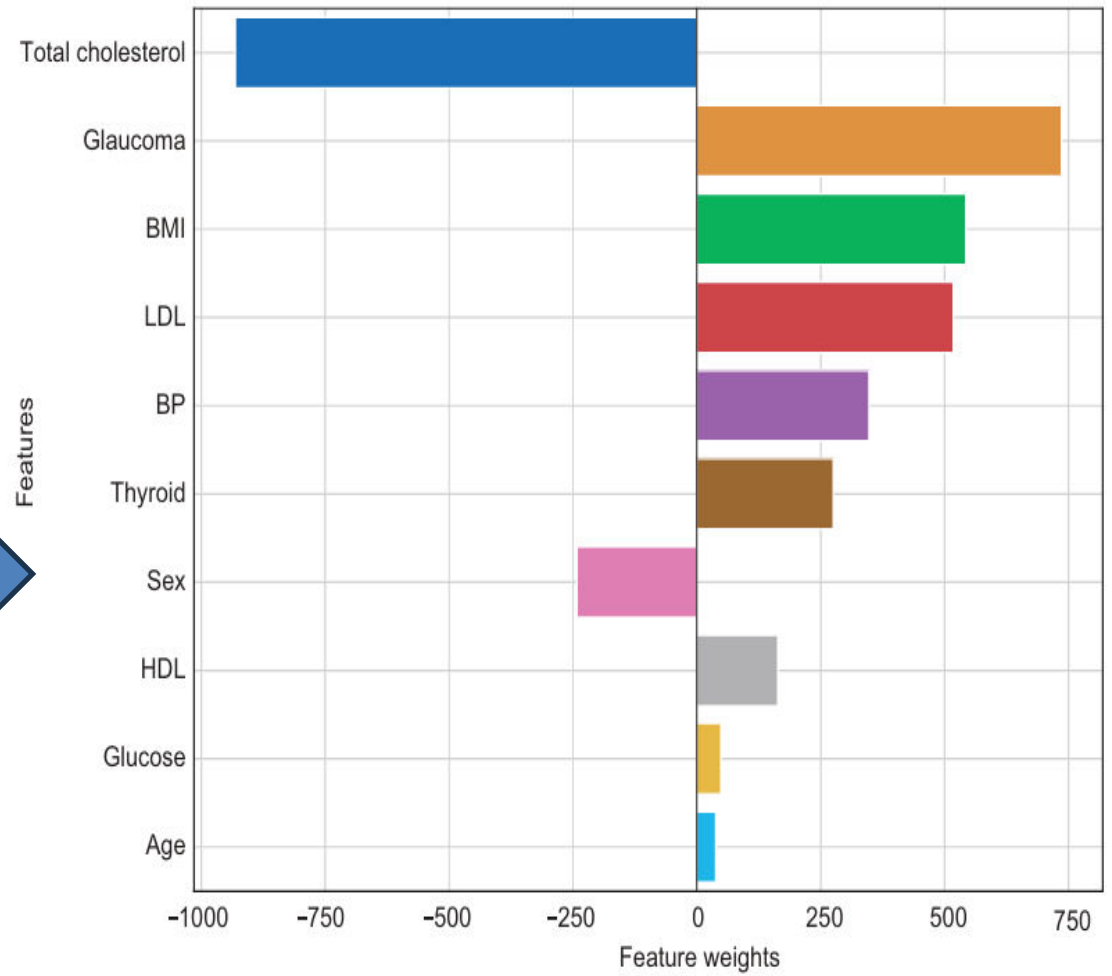
Imports matplotlib and seaborn to plot the feature importance

Obtains the weights from the linear regression model trained earlier using the coef_ parameter

Sorts the weights in descending order of importance and gets their indices

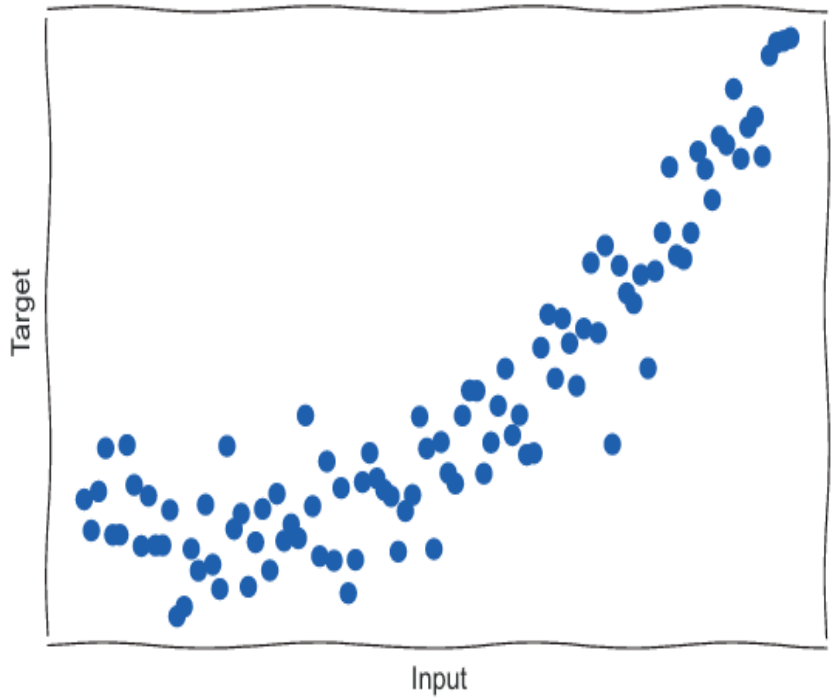
Uses the ordered indices to get the feature names and the corresponding weight values

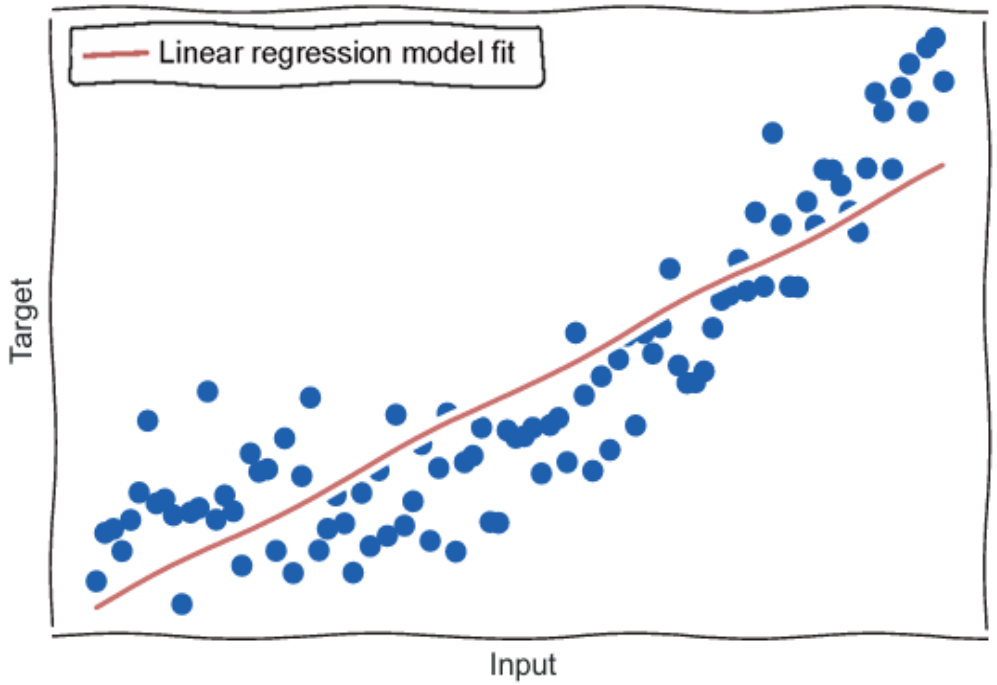
Generates the plot shown in figure 2.7



Limitations of Linear Regression :

• It is highly transparent and easy to understand.
•However, it has poor predictive power, especially in cases where the relationship between the input features and target is **non-linear**





Linear Regression

Use Case 2: House Price Prediction

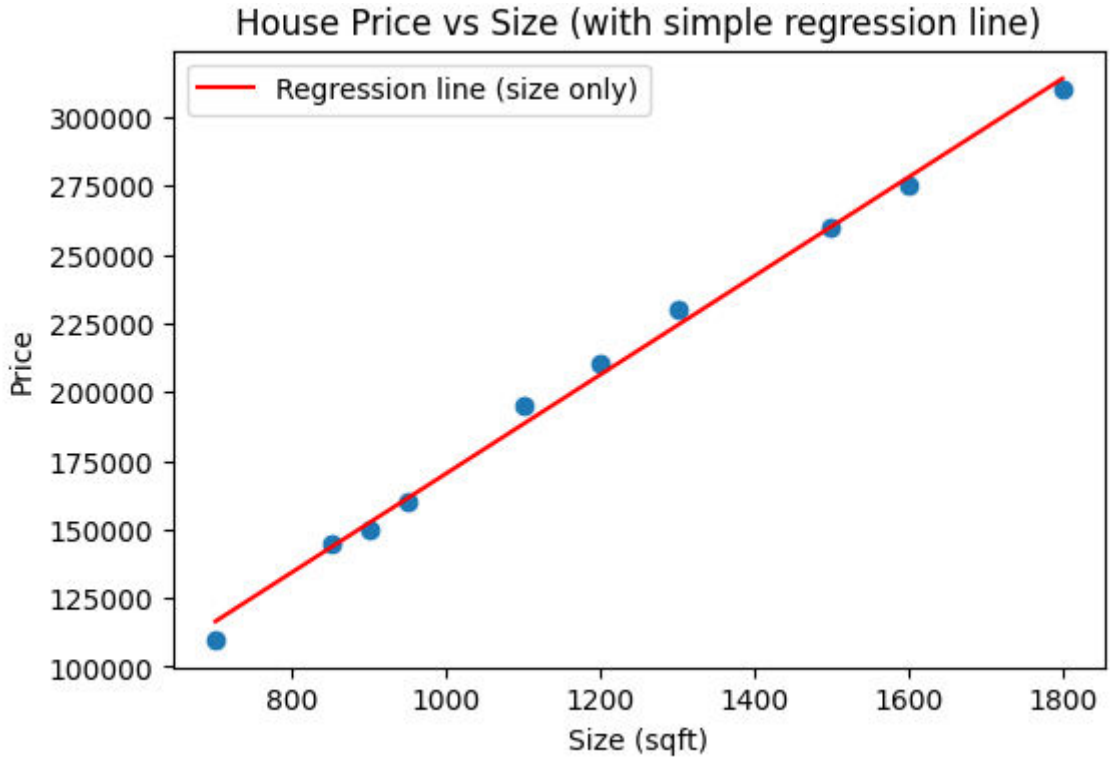
Predicting residential property values using interpretable features that real estate professionals and buyers can understand.

Input Features:

- Square footage
- Neighborhood location score
- Number of bedrooms and bathrooms
- Property age

Output: Predicted price in USD

```
from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(X_train, y_train)

# Interpret coefficients
print(model.coef_)
# [245.3, 18500, 15000, -2300]
# Each sqft adds $245
# Each room adds $15k
```



The coefficient for square footage (245.3) tells us that each additional square foot increases the predicted price by approximately $245, holding all other features constant.

Logistic Regression: Binary Outcomes

Logistic regression extends linear models to classification problems by predicting the probability of binary outcomes. It uses the sigmoid function to transform linear combinations into probabilities between 0 and 1.

Linear Combination

Compute weighted sum of features: $z = \beta_0 + \beta_1 x_1 + \dots$

Sigmoid Transform

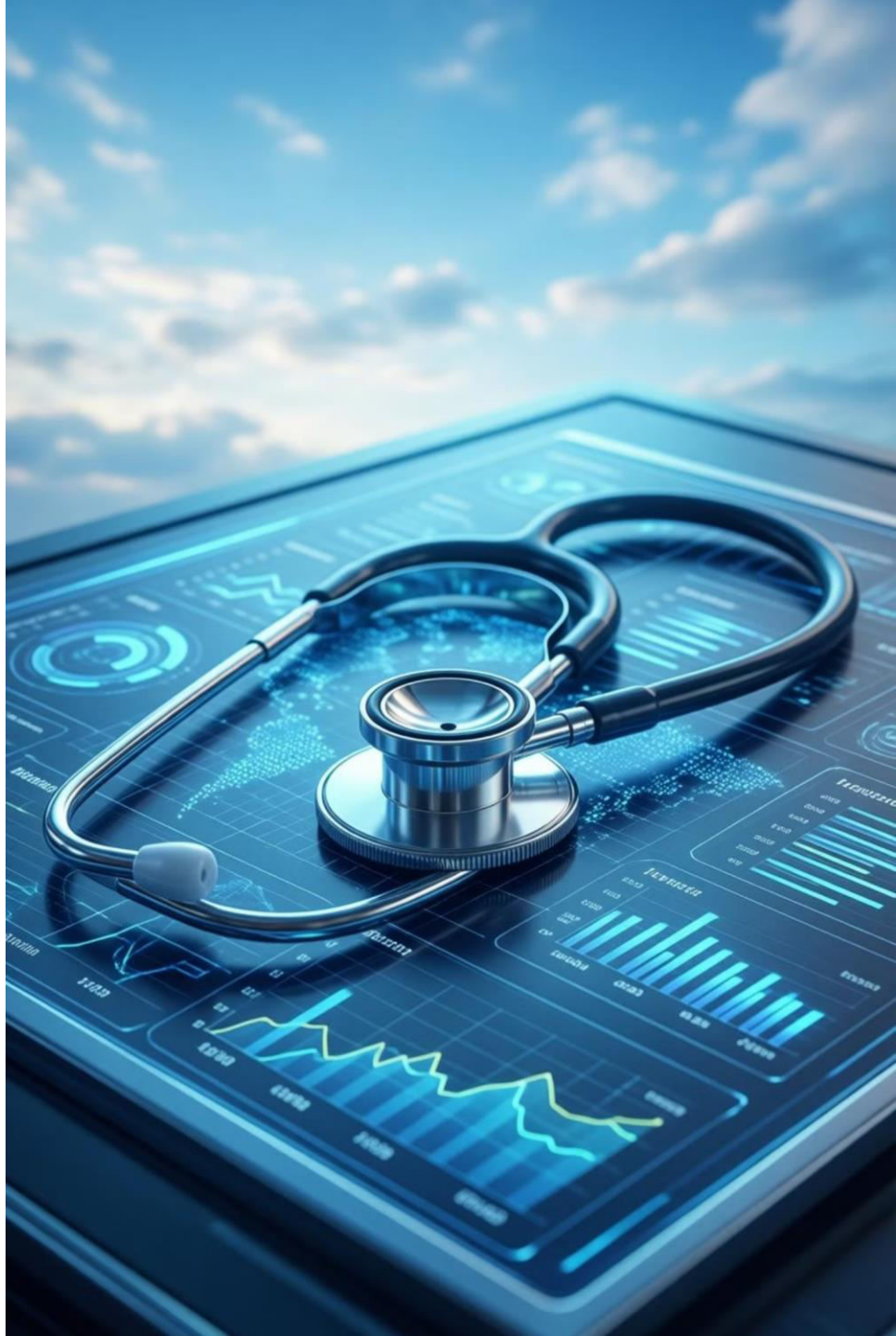
Map to probability space using sigmoid function

Interpret Log-Odds

Coefficients represent change in log-odds for unit increase in feature

$$P(y = 1 \mid x) = \frac{1}{1 + e^{-z}}$$

The sigmoid function ensures outputs are valid probabilities



Logistic Regression Example

Disease Risk Prediction:

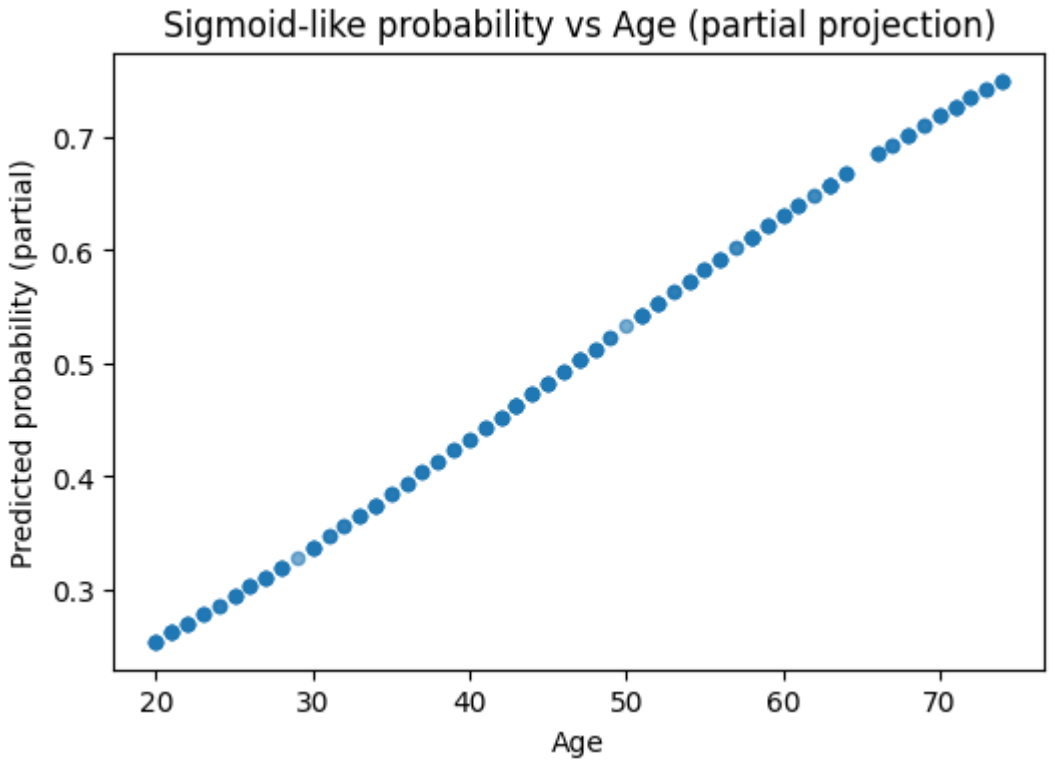
Estimating the probability of disease onset based on patient health indicators. This application is common in clinical decision support systems.

Patient Features:

- Age (years)
- Body Mass Index (BMI)
- Systolic blood pressure
- Family history (binary)

Output: Probability of disease (0 to 1)

```
from sklearn.linear_model import LogisticRegression

model = LogisticRegression()
model.fit(X_train, y_train)

# Coefficients as log-odds
print(model.coef_)
# [[0.05, 0.12, 0.03, 0.89]]

# Predict probability
prob = model.predict_proba(X_test)
print(prob[:, 1])  # P(disease)
```



A coefficient of 0.05 for age means each additional year increases the log-odds of disease by 0.05, corresponding to a $e^{0.05} \approx 1.05$ multiplicative increase in odds.

Generalized Linear Models (GLM)

GLMs extend the linear modeling framework by introducing link functions that connect the linear predictor to the response variable. This flexibility enables modeling diverse data types while maintaining interpretability.

Input Features

Predictor variables X

Link Function

Transform: $g(\mu) = X\beta$

Predicted Output

Expected value μ

Distribution Family

Exponential family (normal, binomial, Poisson)

Common Link Functions

- **Logit:** Binary classification
- **Log:** Count data (Poisson regression)
- **Identity:** Continuous outcomes (standard linear)
- **Probit:** Alternative for binary outcomes

Ideal Applications

- Insurance claim frequency modeling
- Epidemiological count data
- Credit risk assessment
- Quality control defect rates

Key Components Contributing to GLM Interpretability:

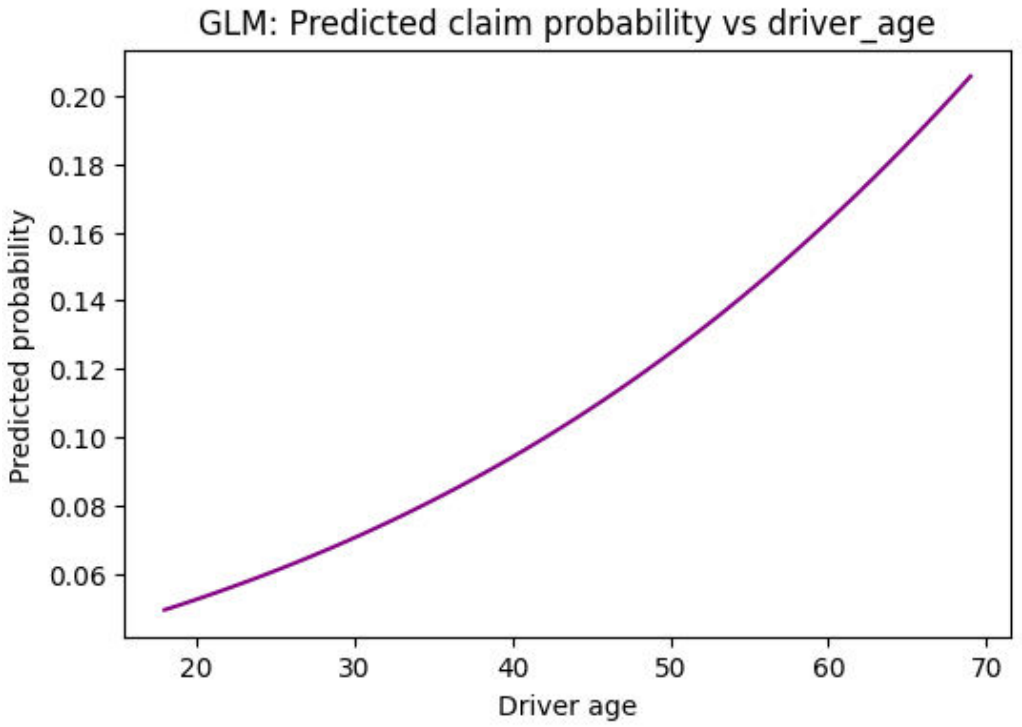
- **Linear Predictor**: The core of a GLM is a linear combination of input features and their corresponding coefficients

(β weights)).

- **Coefficients (Weights)**: The magnitude and sign of the coefficients directly indicate the direction and strength of each feature's impact on the predicted outcome. For example, in a logistic regression (a type of GLM), a positive coefficient means that increasing the feature's value increases the probability of the positive class.
- **Link Function**: This function connects the linear predictor to the expected value of the response variable (e.g., using a logit link for probabilities between 0 and 1, or a log link for positive count data). By inverting the link function, the effect of features on the expected outcome can be directly interpreted (e.g., a one-unit increase in a feature multiplies the expected count by a specific factor).

- **Assumed Distribution**: GLMs require specifying a probability distribution (from the exponential family, such as Gaussian, Poisson, or Binomial) for the target variable. This assumption is transparent and guides how the model should be interpreted in context.

GLM Example: Insurance Risk



Predicting Claim Probability

Insurance companies use GLMs with binomial family and logit link to model the probability of claims, enabling accurate premium pricing and risk stratification.

```
import statsmodels.api as sm

# Binomial family with logit link
glm = sm.GLM(y, X,
    family=sm.families.Binomial())

result = glm.fit()
print(result.summary())

# Coefficient interpretation
# exp(coef) = odds ratio
```

Model Features:

- Driver age and experience
- Vehicle type and age
- Claims history
- Geographic region

Why GLM?

The logit link naturally handles probability bounds (0-1) while binomial distribution matches the binary claim/no-claim outcome structure.

Generalized Linear Model Regression Results

Dep. Variable:	claim	No. Observations:	400
Model:	GLM	Df Residuals:	396
Model Family:	Binomial	Df Model:	3
Link Function:	Logit	Scale:	1.0000
Method:	IRLS	Log-Likelihood:	-138.34
Date:	Mon, 12 Jan 2026	Deviance:	276.68
Time:	17:30:21	Pearson chi2:	414.
No. Iterations:	5	Pseudo R-squ. (CS):	0.05071
Covariance Type:	nonrobust		

	coef	std err	z	P>|z|	[0.025	0.975]
const	-5.5266	0.915	-6.039	0.000	-7.320	-3.733
driver_age	0.0315	0.011	2.909	0.004	0.010	0.053
annual_mileage	0.0001	5.32e-05	2.752	0.006	4.21e-05	0.000
prior_accidents	0.4349	0.209	2.084	0.037	0.026	0.844

Generalized Additive Models (GAM)

GAMs maintain an additive structure like linear models but allow each feature to have its own flexible, smooth function. This captures non-linear relationships while preserving interpretability through visualization of individual feature effects.

$$y = \beta_0 + f_1(x_1) + f_2(x_2) + f_3(x_3) + \cdots + f_n(x_n)$$

Each f_i is a smooth, non-parametric function learned from data

Non-Linear Flexibility

Captures complex patterns like U-shapes, plateaus, and thresholds that linear models miss.

Visual Interpretability

Plot each feature's partial effect to see exactly how it influences predictions across its range.

Additive Structure

Effects combine additively, making it easy to understand individual feature contributions.

GAM Example: Customer Churn

Predicting customer churn with GAMs reveals non-linear relationships that business stakeholders can visualize and act upon. For instance, churn risk might decrease rapidly at first with tenure, then plateau after 12 months.

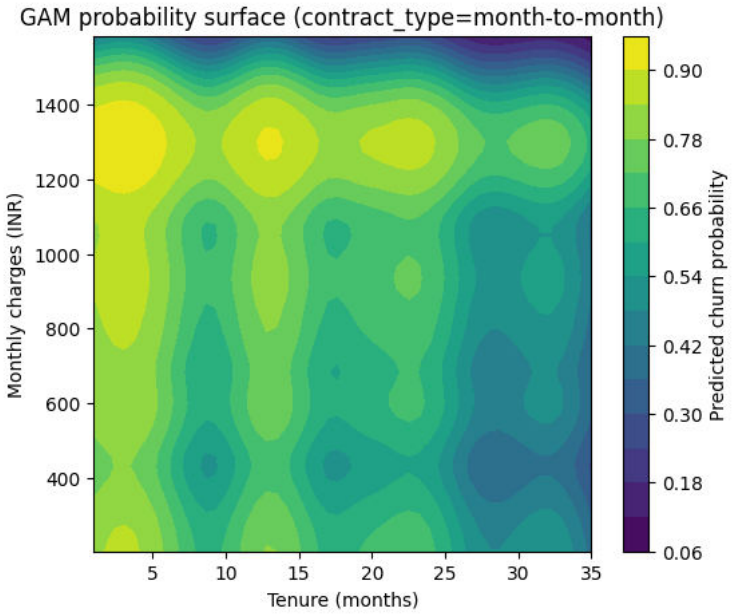
Key Features:

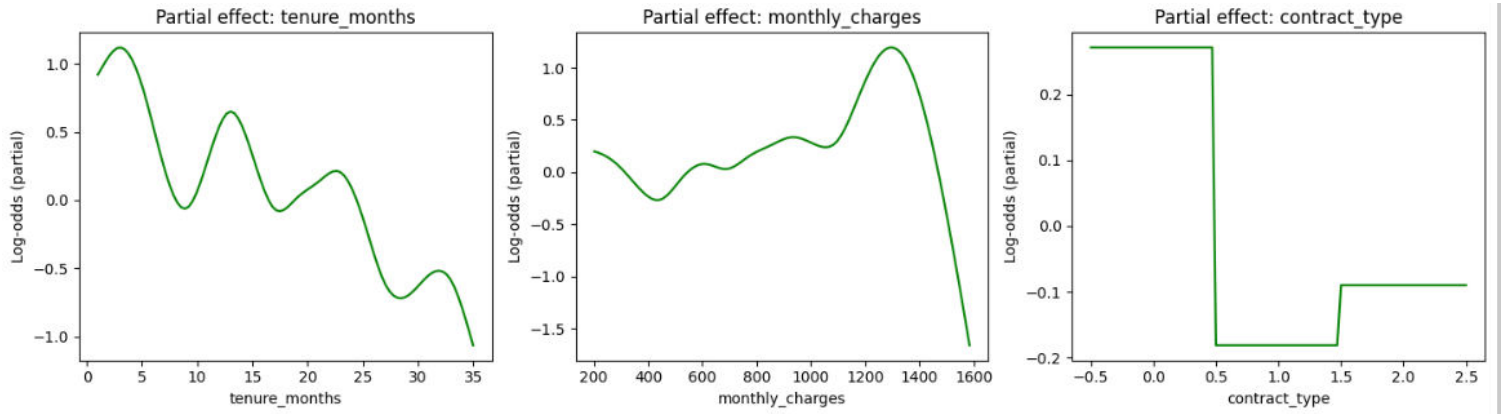
- Customer tenure (months)
- Monthly charges ($)
- Contract type (ordinal)
- Support tickets (count)

Interpretable Insights:

The smooth curves reveal actionable patterns—churn spikes when monthly monthly charges exceed $80, but long-term contracts show protective effects protective effects regardless of price.

```
GAM summary:
LogisticGAM
============================================= ==========================================================
Distribution:                      BinomialDist Effective DoF:                                     19.8747
Link Function:                        LogitLink Log Likelihood:                                    -307.165
Number of Samples:                          500 AIC:                                               654.0795
                                                AICc:                                              655.9896
                                                UBRE:                                                 3.34
                                                Scale:                                                 1.0
                                                Pseudo R-Squared:                                    0.0749
==========================================================================================================
Feature Function                  Lambda               Rank         EDoF         P > x        Sig. Code
================================= ==================== ============ ============ ============ ============
s(0)                              [0.6]                20           10.4         4.05e-04     ***
s(1)                              [0.6]                20           7.6          7.91e-01
f(2)                              [0.6]                3            1.9          2.02e-01
intercept                                              1            0.0          7.32e-01
==========================================================================================================
Significance codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

WARNING: Fitting splines and a linear function to a feature introduces a model identifiability problem
         which can cause p-values to appear significant when they are not.

WARNING: p-values calculated in this manner behave correctly for un-penalized models or models with
         known smoothing parameters, but when smoothing parameters have been estimated, the p-values
         are typically lower than they should be, meaning that the tests reject the null too readily.
None
/tmp/ipython-input-325674001.py:16: UserWarning: KNOWN BUG: p-values computed in this summary are likely much smaller than they should be.

Please do not make inferences based on these values!

Collaborate on a solution, and stay up to date at:
github.com/dswah/pyGAM/issues/163
```





THANK YOU

Elective - III

UE23AM342BA3 : Explainable AI

Decision Tree and Rule Based Methods

Prof.Ayisha Noori V K

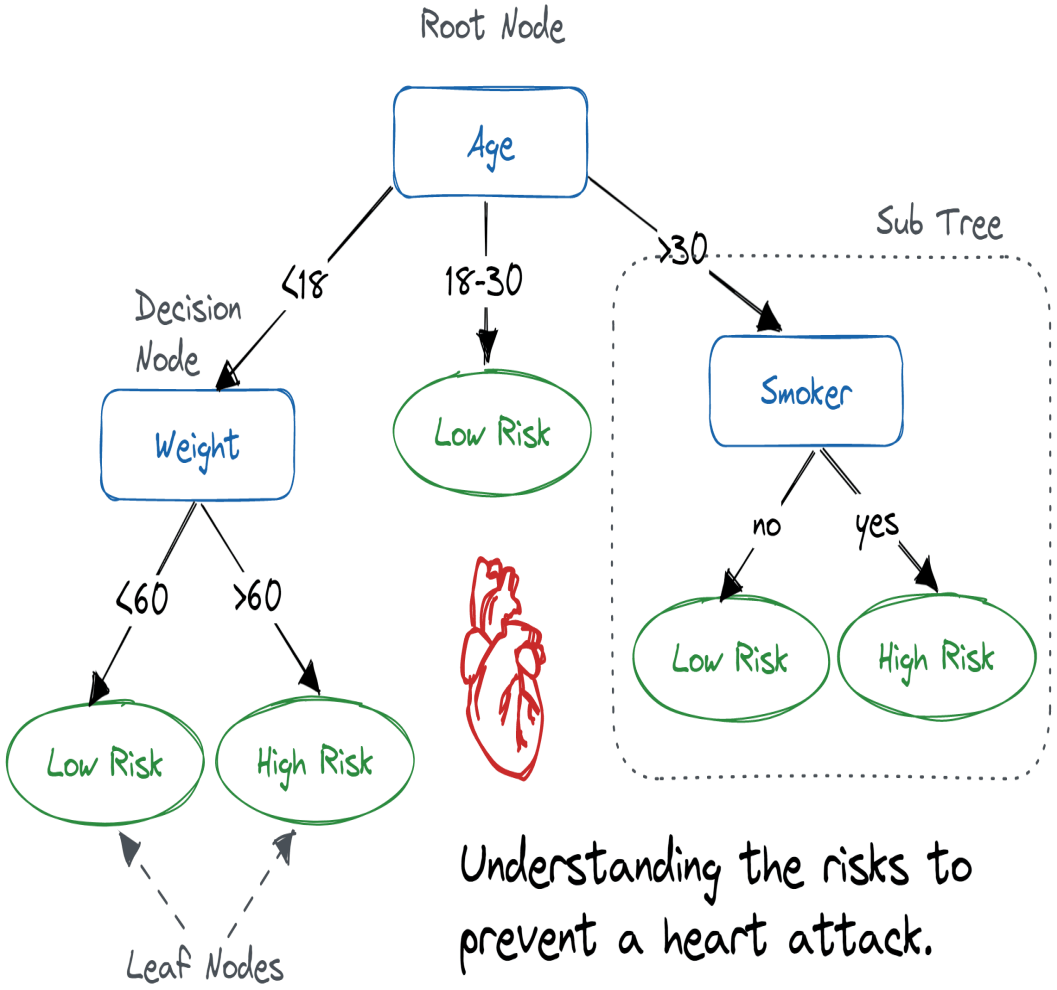
Department of Computer Science and Engineering(AI&ML)

Contents:

- Decision Tree
- Rule Based Methods
- Anchors
- Counter factual explanations
- Structured Data Explainers
- Recap of Unit1

DECISION TREE:

- A decision tree is a great machine learning algorithm that can be used to model complex nonlinear relationships. It can be applied to both regression and classification tasks.
- It has relatively higher predictive power than linear regression and is highly interpretable, too
- The basic idea behind a decision tree is to find optimum splits in the data that best predict the output or target variable.



Decision Tree (Disease Risk Dataset)

Concept: Decision Tree Classifier

Dataset: disease_risk.csv (age, BMI, systolic BP, disease)

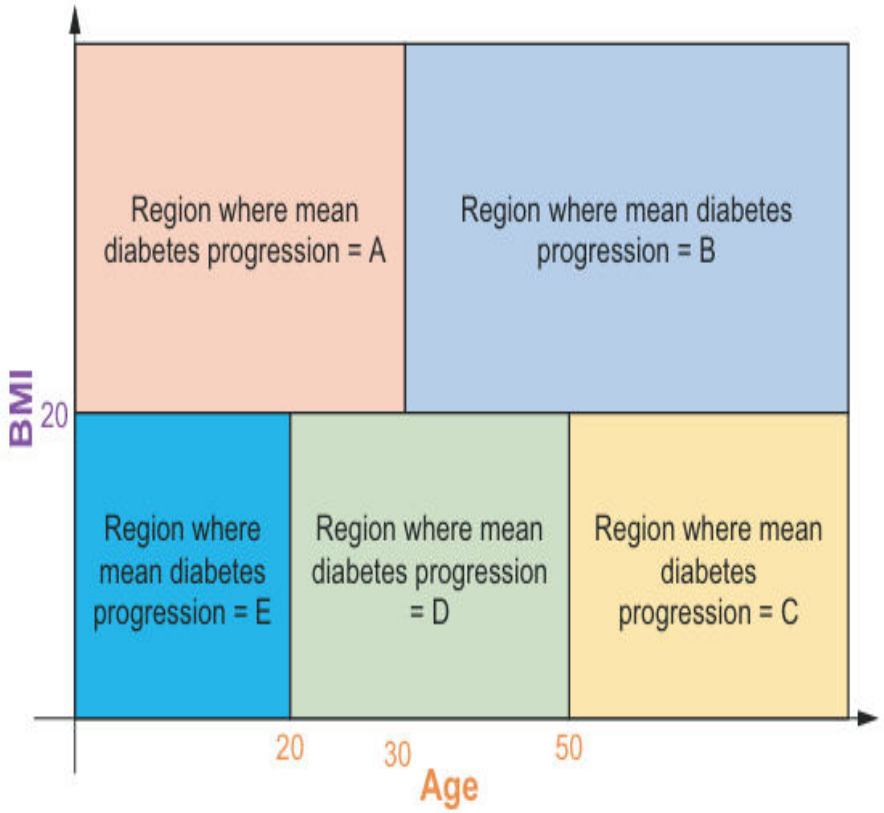
Explanation: Decision Trees split data into branches based on feature thresholds. Each branch leads to a prediction about disease risk.

Visualization: Tree diagram showing splits on age, BMI, and blood pressure.

Layman's View: Like a doctor's flowchart: "If age > 50 and BP > 130 → higher risk."

Decision Tree: Case Study

- Consider the previous diabetic example by considering only two features, BMI and Age. The decision tree splits the dataset into five groups in total, three age groups and two BMI groups



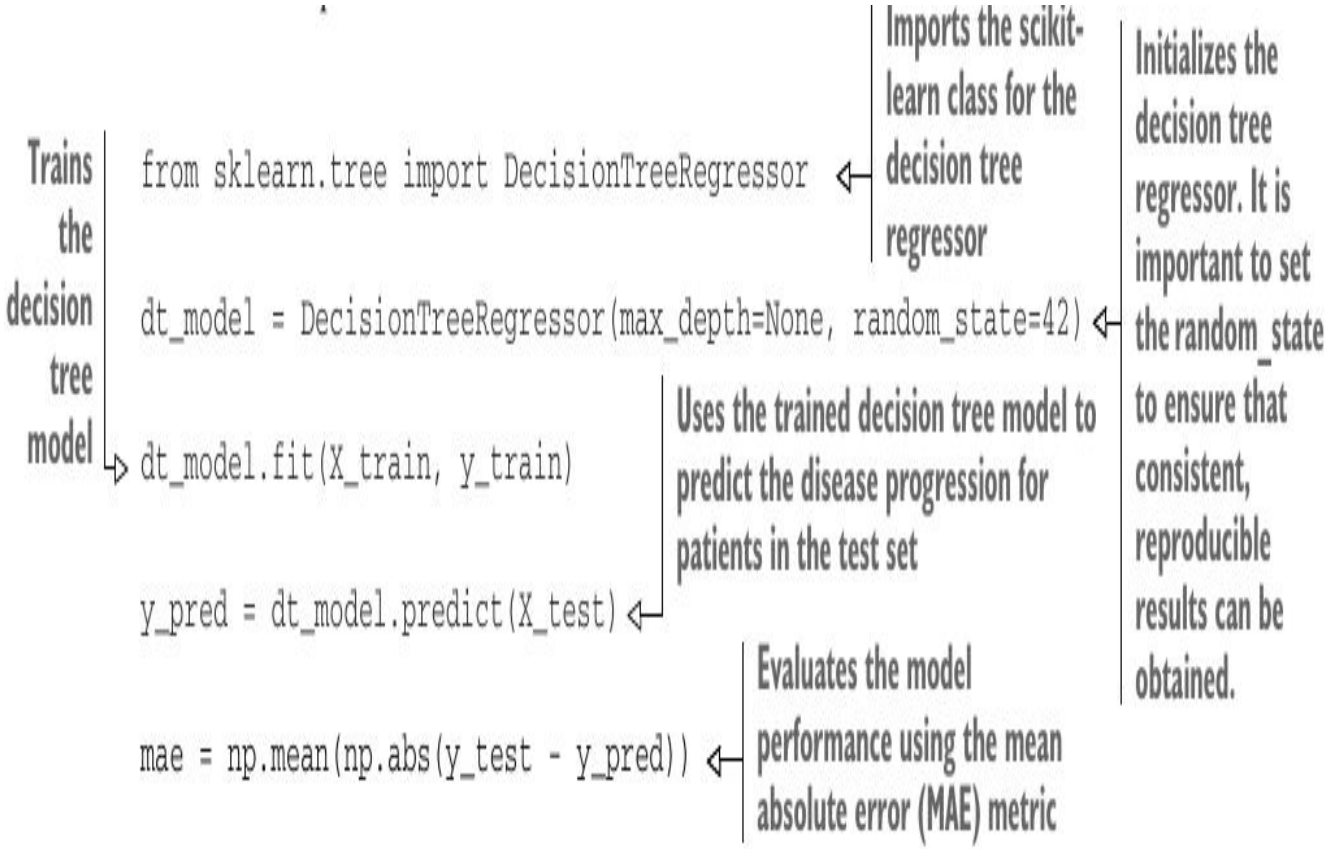
- The algorithm that is commonly applied in determining the optimum splits is the **classification and regression tree (CART)** algorithm.
- This algorithm first chooses a feature and a threshold for that feature. Based on that feature and threshold, the algorithm splits the dataset into the following two subsets:
  - **Subset 1, where the value of the feature is less than or equal to the threshold**
  - **Subset 2, where the value of the feature is greater than the threshold**

The algorithm picks the feature and threshold that minimizes a cost function or criterion. For **regression tasks**, this criterion is typically the **mean squared error (MSE)**, and for **classification tasks**, it is typically either **Gini impurity or entropy**.

The algorithm then continues to recursively split the data until the criterion is reduced further or until a maximum depth is reached

Case Study : Implementation as Binary Tree

- A decision tree model can be trained in Python using the Scikit-Learn package as follows. The code to learn the open diabetes dataset and to split it into the training and test sets is the same as the one used for linear regression



Decision Tree for Classification

```
from sklearn.tree import DecisionTreeClassifier
dt_model = DecisionTreeClassifier(criterion='gini',
max_depth=None)
dt_model.fit(X_train, y_train)
```

The criterion parameter in the DecisionTreeClassifier can be used to specify the cost function for the CART algorithm. By default, it is set to gini, but it can be changed to entropy

Visualization of the Diabetes Decision Tree Model:

```
from sklearn.externals.six import StringIO
from IPython.display import Image
from sklearn.tree import export_graphviz
import pydotplus

diabetes_dt_dot_data = StringIO()
export_graphviz(dt_model,
                out_file=diabetes_dt_dot_data,
                filled=False, rounded=True,
                feature_names=feature_names,
                proportion=True,
                precision=1,
                special_characters=True)
dt_graph = pydotplus.graph_from_dot_data(diabetes_dt_dot_data.getvalue())
Image(dt_graph.create_png())
```

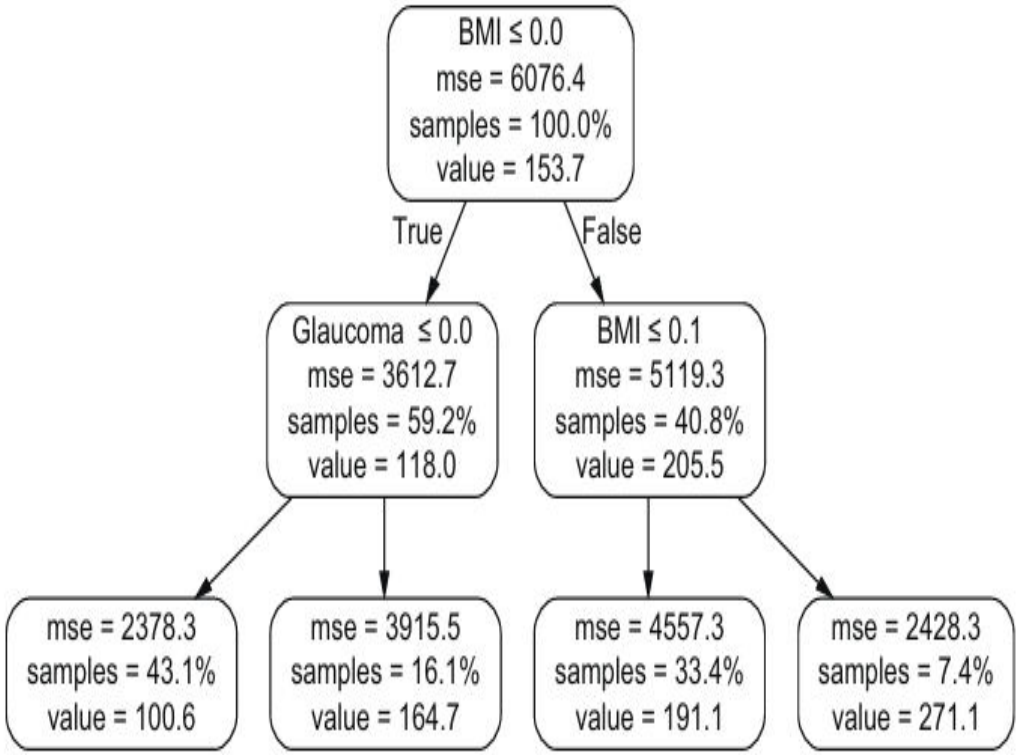
Imports all the necessary libraries to generate and visualize the binary tree

Initializes a string buffer to store the binary tree/graph in DOT format

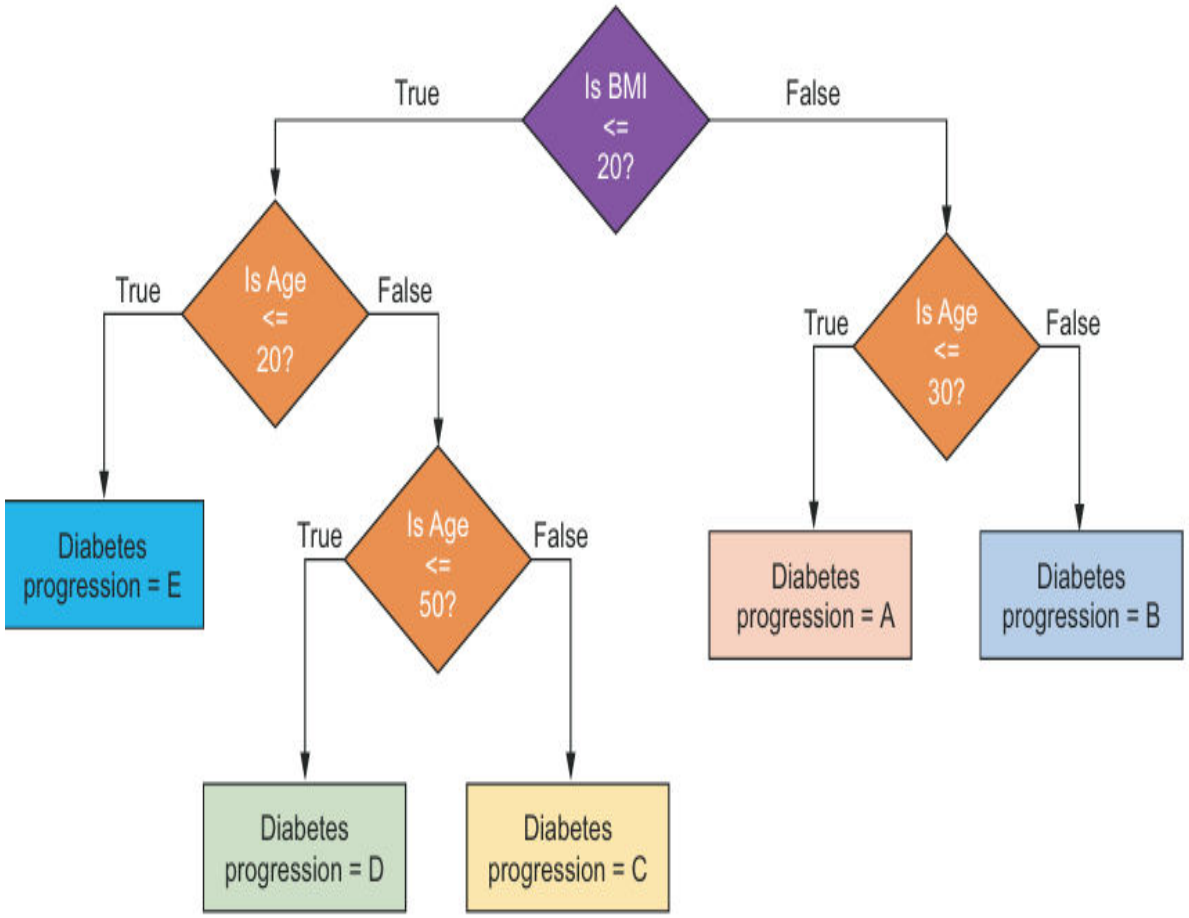
Exports the decision tree model as a binary tree in DOT format

Generates an image of the binary tree using the DOT format string

Visualizes the binary tree using the Image class



Interpreting Decision Trees :



A decision tree can be visualized as a bunch of if-else conditions strung together, where each condition splits the data in two. Such a model can be easily visualized as a binary tree. The tree can be inter preted as follows.

Starting at the root of the tree, check if the normalized BMI is <= 0. If true, go to the left part of the tree. If false, go to the right part of the tree. Because we are starting at the root of the tree, this node accounts for 100% of the data. This is why samples is equal to 100%. Also, if we were to set the max_depth to 0 and predict the disease progression, then we would use the average value of all the samples in the data, which is 153.7, represented as value in the tree. By predicting 153.7, we would get an MSE of 6076.4.

If the normalized BMI is <= 0, then we go to the left part of the tree and check if the normalized Glaucoma is <= 0. If BMI is <= 0, we would account for approximately 59% of the data, and the MSE would reduce from 6076.4 for the parent node to 3612.7. We can repeat this process until we have reached the leaf nodes in the tree. If we look at, say, the right-most leaf node, this corresponds to the following condition: if BMI > 0 and BMI > 0.1 and LDL > 0, then predict 225.8 for 2.3% of the data, resulting in an MSE of 2757.9.

The complexity of this tree will increase as max_depth increases or as the number of input feature increases

Computing Feature importance:

To compute the feature importance, we first need to compute the importance of a node in the binary tree. The importance of a node is computed as the decrease in the cost function or impurity measure for that node weighted by the probability of reaching that node in the tree. Python code is given below:

$$I_k^{node} = p_k \cdot m_k - p_k^{(left)} \cdot m_k^{(left)} - p_k^{(right)} \cdot m_k^{(right)}$$

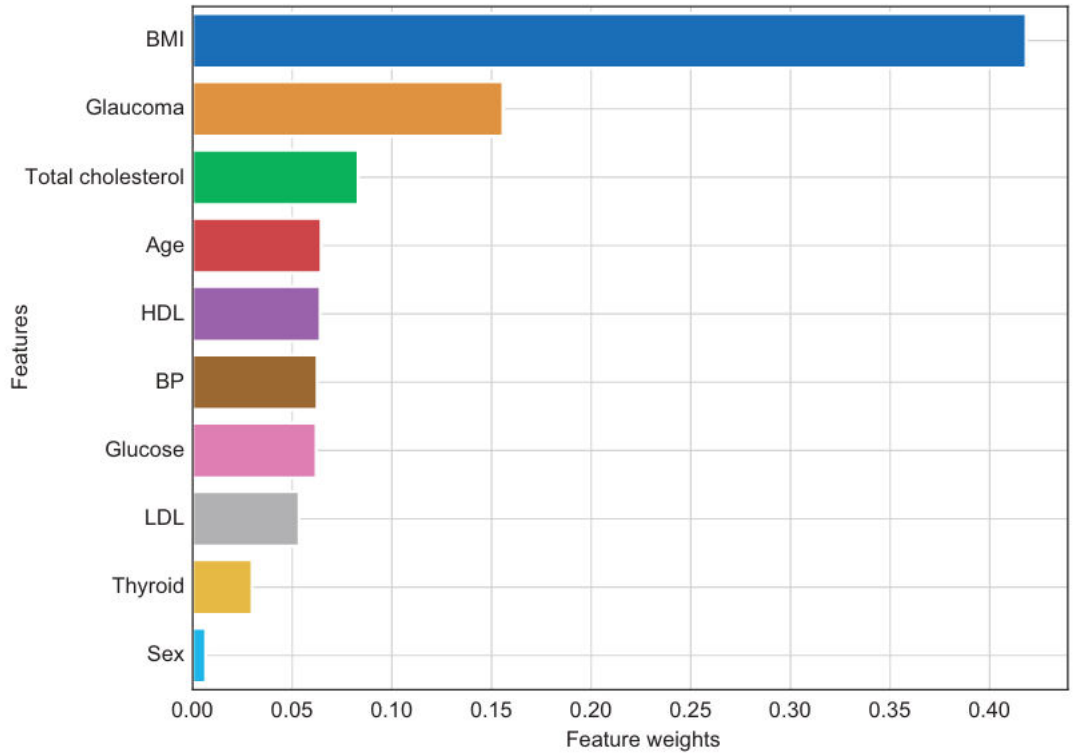
where I_k^{node} = Importance of node k, p_k = Proportion of samples to reach node k, m_k = Impurity measure of node k, $p_k^{(left)}$ = Proportion of samples to reach left subtree of node k, $m_k^{(left)}$ = Impurity measure of left subtree of node k, $p_k^{(right)}$ = Proportion of samples to reach right subtree of node k, $m_k^{(right)}$ = Impurity measure of right subtree of node k.

$$I_i^{feature} = \frac{\sum_{j \in \mathbb{J}} I_j^{node}}{\sum_{k \in \mathbb{K}} I_k^{node}}$$

where $I_i^{feature}$ = Importance of feature i, $\sum_{j \in \mathbb{J}} I_j^{node}$ = Sum of importance of all nodes j that split on feature i, $\sum_{k \in \mathbb{K}} I_k^{node}$ = Sum of importance of all nodes k in the decision tree

Imp

```
# Gets feature importance from the trained decision tree model
weights = dt_model.feature_importances_

# Sorts indices of feature weights in descending order of importance
feature_importance_idx = np.argsort(np.abs(weights))[::-1]
# Gets the feature names and feature weights in descending order of importance
feature_importance = [feature_names[idx].upper() for idx in
  feature_importance_idx]
feature_importance_values = [weights[idx] for idx in
  feature_importance_idx]

# Generates the plot shown in figure 2.14
f, ax = plt.subplots(figsize=(10, 8))
sns.barplot(x=feature_importance_values, y=feature_importance, ax=ax)
ax.grid(True)
ax.set_xlabel('Feature Weights')
ax.set_ylabel('Features')
```



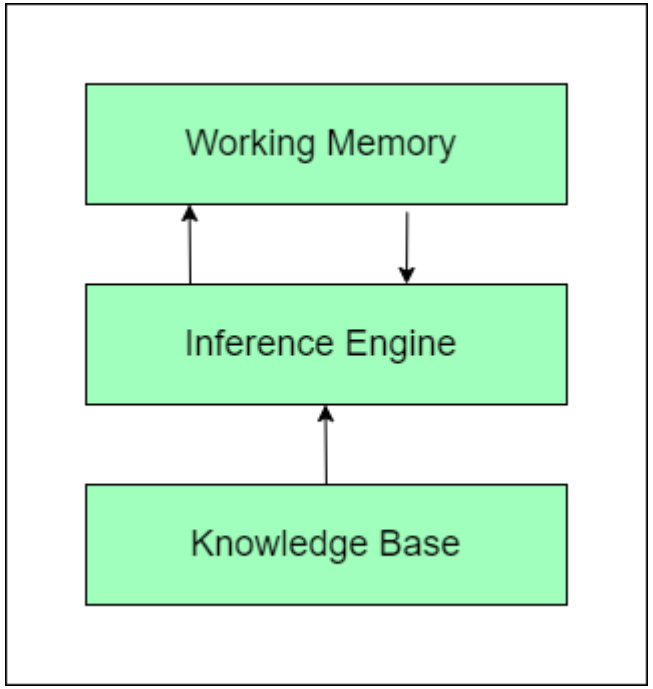
RULE BASED METHODS:

• Rule-based methods are a core Explainable AI (XAI) approach that use explicit "if-then" logic to make transparent, human-understandable decisions, forming intrinsically interpretable models like Decision Trees or acting as simpler "**surrogate**" models to explain complex "black-box" systems

•They provide clarity by showing exactly which conditions triggered a conclusion, making them ideal for critical domains like healthcare and finance where tracing decisions is vital

There are many ways to learn rules from data:

1. **OneR** : learns rules from a single feature. OneR is characterized by its simplicity, interpretability and its use as a benchmark. A OneR model is a decision tree with only one split. The split is not necessarily binary as in CART, but depends on the number of unique feature values
2. **Sequential covering :** is a general procedure that iteratively learns rules and removes the data points that are covered by the new rule. This procedure is used by many rule learning algorithms.
3. **Bayesian Rule Lists:** combine pre-mined frequent patterns into a decision list using Bayesian statistics. Using pre-mined patterns is a common approach used by many rule learning algorithms



ANCHORS(Scoped Rules) :

- In Explainable AI (XAI), Anchors are high-precision, model-agnostic IF-THEN rules that identify sufficient conditions (the "anchor") for a model's prediction, ensuring that if the anchor conditions hold, the prediction stays the same, regardless of other feature changes, making explanations clear and reusable for various data types like text, image, and tabular data
- Anchors utilizes reinforcement learning techniques in combination with a graph search algorithm to reduce the number of model calls (and hence the required runtime) to a minimum while still being able to recover from local optima
- This method is therefore considered "**Local**", "**Post hoc**" and "**Model agnostic**"

How Anchors Work:

- **Perturbation-Based:** Like LIME, Anchors perturb the input data to find rules.
- **Rule Generation:** They use reinforcement learning and graph search to find minimal sets of feature conditions (predicates) that, when met, lead to a specific high-precision outcome.
- **High Precision & Coverage:** An anchor rule is a set of conditions (e.g., "IF age > 40 AND income < 50k") that strongly predicts an outcome, with "coverage" indicating how many instances this rule applies to.

Anchor : Example

For instance, suppose we are given a bivariate black box model that predicts whether or not a passenger survived the Titanic disaster. Now we would like to know why the model predicts for one specific person (see Table) that they survived. The anchors algorithm provides a result explanation like the one shown below.

Feature	Value
Age	20
Sex	female
Class	first
Ticket price	300$
More attributes	...
Survived	true

And the corresponding anchors explanation is:

IF SEX = female AND Class = first THEN PREDICT Survived = true WITH PRECISION 97% AND COVERAGE 15%

An anchor A is formally defined as follows:

$$\mathbb{E}_{\mathcal{D}_{\mathbf{x}}(\mathbf{z}|A)}[1_{\hat{f}(\mathbf{x})=\hat{f}(\mathbf{z})}] \geq \tau; A(\mathbf{x}) = 1$$

Given an instance $\mathbf{x}$ to be explained, a rule or an anchor A is to be found, such that it applies to $\mathbf{x}$, while the same class as for $\mathbf{x}$ gets predicted for a fraction of at least τ of $\mathbf{x}$'s neighbors where the same A is applicable. A rule's precision results from evaluating neighbors or perturbations (following $\mathcal{D}_{\mathbf{x}}(\mathbf{z}|A)$) using the provided machine learning model (denoted by the indicator function $1_{\hat{f}(\mathbf{x})=\hat{f}(\mathbf{z})}$).

Anchor example

Feature	Value
Age	20
Sex	female
Class	first
Ticket Price	300$
More attributes	...
Survived	**true**

IF `SEX = female`
AND `Class = first`
THEN PREDICT `Survived = true`
WITH PRECISION `97%`
AND COVERAGE `15%`

Anchors : Formal Definition

An anchor A is formally defined as follows:

$$\mathbb{E}_{\mathcal{D}_{\mathbf{x}}(\mathbf{z}|A)}[1_{\hat{f}(\mathbf{x})=\hat{f}(\mathbf{z})}] \geq \tau; A(\mathbf{x}) = 1$$

Wherein:

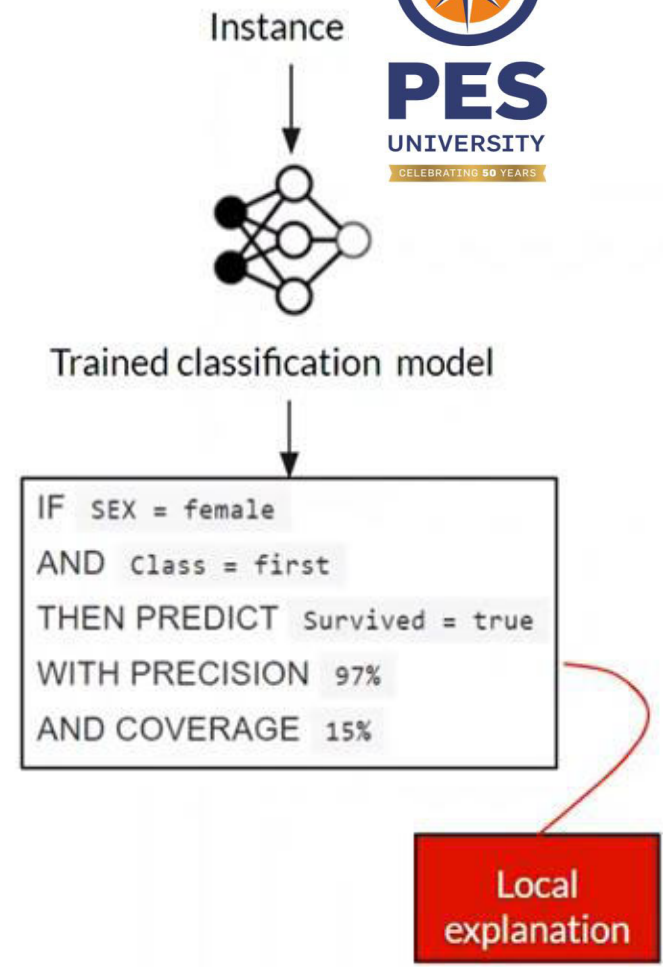
- $\mathbf{x}$ represents the instance being explained (e.g. one row in a tabular dataset).
- A is a set of predicates, i.e., the resulting rule or anchor, such that $A(\mathbf{x}) = 1$ when all feature predicates defined by A correspond to $\mathbf{x}$'s feature values.
- $\hat{f}$ denotes the classification model to be explained (e.g. an artificial neural network model). It can be queried to predict a label for $\mathbf{x}$ and its perturbations.
- $\mathcal{D}_{\mathbf{x}}(\cdot|A)$ indicates the distribution of neighbors of $\mathbf{x}$, matching A.
- $0 \leq \tau \leq 1$ specifies a precision threshold. Only rules that achieve a local fidelity of at least τ are considered a valid result.

Given an instance $\mathbf{x}$ to be explained, a rule or an anchor A is to be found, such that it applies to $\mathbf{x}$, while the same class as for $\mathbf{x}$ gets predicted for a fraction of at least τ of $\mathbf{x}$'s neighbors where the same A is applicable. A rule's precision results from evaluating neighbors or perturbations (following $\mathcal{D}_{\mathbf{x}}(\mathbf{z}|A)$) using the provided machine learning model (denoted by the indicator function $1_{\hat{f}(\mathbf{x})=\hat{f}(\mathbf{z})}$).

Reference : 16 Scoped Rules (Anchors) – Interpretable Machine Learning

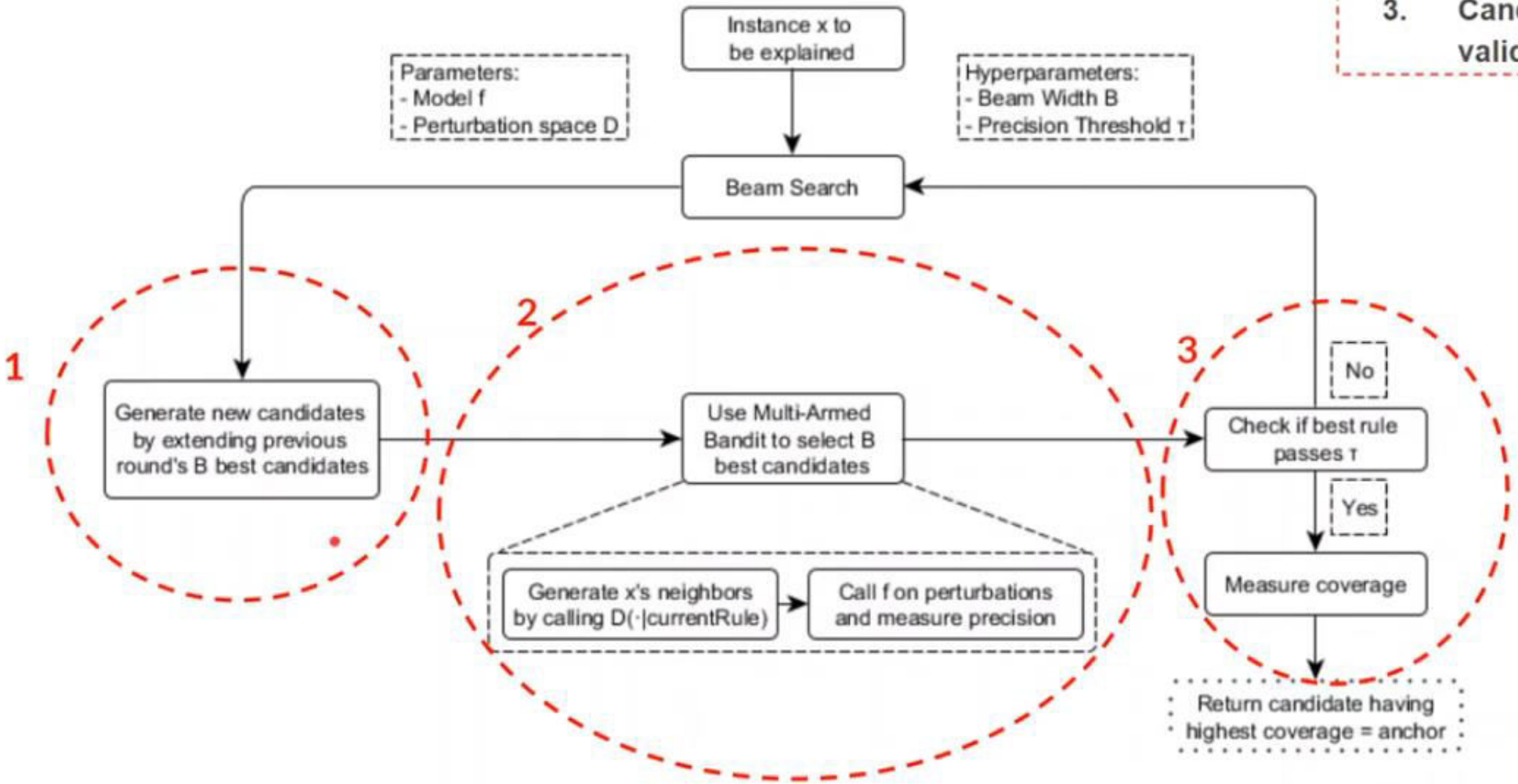
Anchors (Scoped Rules)

- **What is?**
  - Anchors is a model-agnostic method that explains individual predictions
- **How does it work?**
  - It uses a perturbation-based strategy to generate local explanations
- **What is the input?**
  - Instance x to be explained, trained black-box model
- **What is the output?**
  - Easy to understand IF-THEN rules (called Anchors) with precision and coverage
- **What techniques does it use?**
  - Data perturbation (same as LIME)
  - Reinforcement learning (Multi-armed bandit)
  - Graph search algorithm (Beam search)



Anchors algorithm's components

1. Candidate Generation
2. Best Candidate identification
3. Candidate precision validation



Anchor – Example Implementation

To test the Anchor method (in Python), we focus on the “Red Wine Quality ” dataset with physicochemical inputs and **quality** as output

Fixed acidity	Volatile acidity	Citric acid	Residual sugar	Chlorides	Free sulfur dioxide	Total sulfur dioxide	Density	pH	Sulphates	Alcohol	Quality
7,8	0,88	0	2,6	0,098	25	67	0,9968	3,2	0,68	9,8	5
11,2	0,28	0,56	1,9	0,075	17	60	0,998	3,16	0,58	9,8	6
7,4	0,66	0	1,8	0,075	13	40	0,9978	3,51	0,56	9,4	5
7,3	0,65	0	1,2	0,065	15	21	0,9946	3,39	0,47	10	7
7,5	0,5	0,36	6,1	0,071	17	102	0,9978	3,35	0,8	10,5	5

Then, we train a Random Forest model to predict the quality output.

```
import pandas as pd
from sklearn.metrics import accuracy_score
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier

df = pd.read_csv("./winequality-red.csv", sep=';')
feature_columns = df.columns[:-1]
X = df[feature_columns]
y = df['quality']
X_train, X_test, y_train, y_test = train_test_split(X,y,test_size=0.2)
model = RandomForestClassifier(n_estimators=500)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print('Accuracy of random forest: ', accuracy_score(y_test,y_pred))
```

We obtain the following result:

Accuracy of random forest: 0.709375

Use Anchor to better understand your Machine Learning model | by Sia AI | Sia AI | Medium

Anchor – Example Implementation

With that dataset and that trained model, we can apply the Anchor method and we get the following rule for a random point:

```
IF 'alcohol' > 11.10 AND 'total sulfur dioxide' > 22.00
THEN PREDICT 'quality' = 6
WITH Precision = 0.7201 AND Coverage = 0.1432
```

In addition to the generated prediction rule, two metrics are also returned that provide insight about the quality and the relevance of the rule:

- **Precision**: the relative number of correct predictions. In the previous result, the rule was correct 7.2 times on average for 10 predictions.
- **Coverage**: the "perturbation space" (or the anchor's probability of applying to its "neighbors") or the relative number of points generated by the "perturbation" that follow the rule. Generally, the more variables the rule contains, the lower the coverage

Exploration of models with Anchor method:

- From that interpretability method, it is quite possible to explore and understand any prediction model.
- To do so, we will iterate the Anchor method over many points of the dataset.
- The algorithm takes as input a minimal precision threshold for Anchor method and a proportion of points of the dataset implied by all the rules and as output, the list of all the rules.
- About the exploration, the principle is the following:

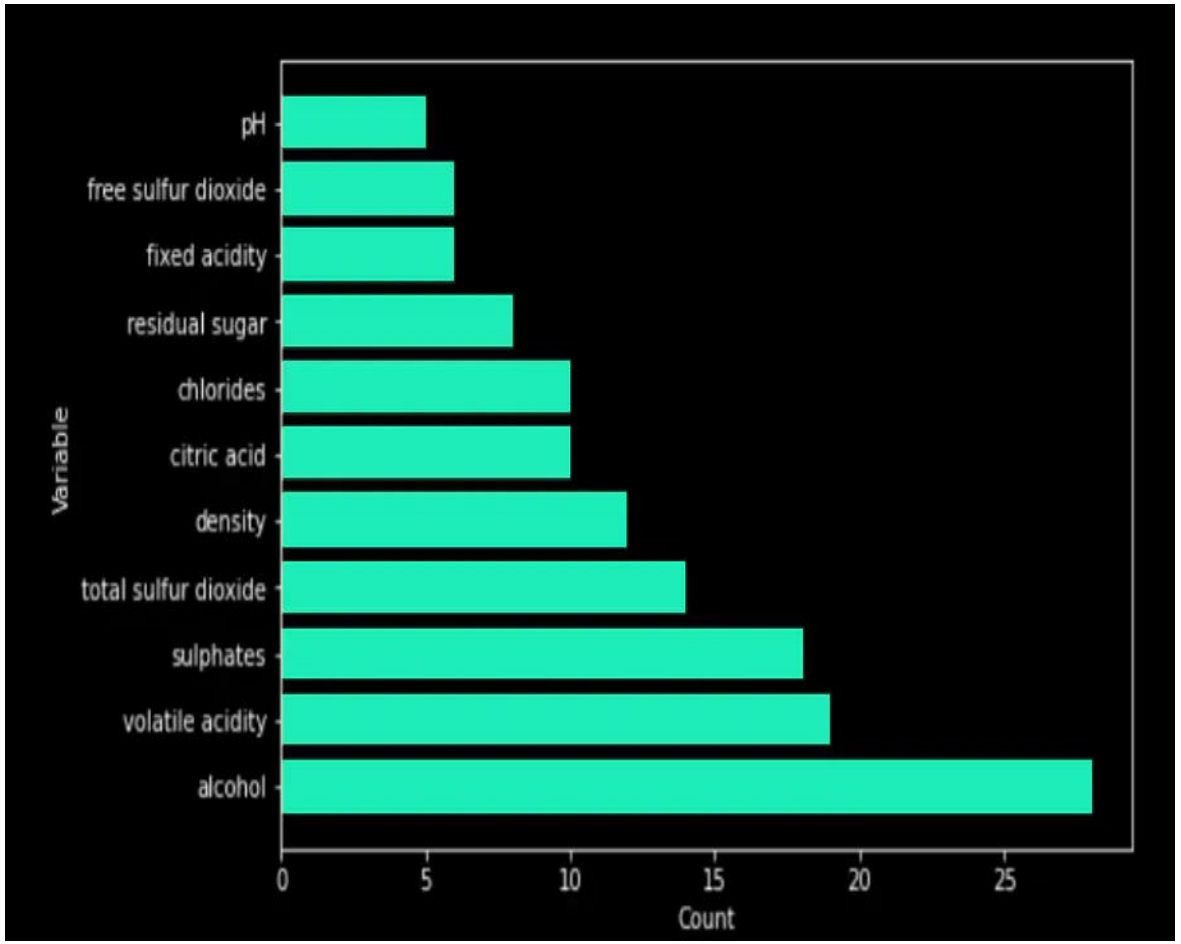
1. Get a random point in the dataset
2. Apply Anchor method on that point
3. Remove the points affected by the rule from the dataset
4. Save the rule
5. Repeat from step 1 until less than 20% of points remain

By applying the exploration algorithm described above with a minimum accuracy threshold of 0.7 and by traversing at least 80% of the dataset, we generate more than thirty prediction rules. Here are some of them

```
IF 'sulphates' > 0.73 AND 'volatile acidity' <= 0.39 AND 'alcohol' > 10.20 AND 'citric acid' <= 0.26
AND ''density' <= 1.00 AND 'pH' <= 3.31 AND 'fixed acidity' > 9.20 AND 'free sulfur dioxide' <= 21.00
AND 'chlorides' <= 0.08
THEN PREDICT 'quality' = 7
WITH Precision = 0.719 AND Coverage = 0.005
IF 'alcohol' > 10.20 AND 'residual sugar' <= 2.20 AND 'density' > 1.00 AND 'volatile acidity' <= 0.39
THEN PREDICT 'quality' = 6
WITH Precision = 0.762 AND Coverage = 0.006
IF 'alcohol' > 11.10 AND 'sulphates' <= 0.73 AND 'volatile acidity' <= 0.52
THEN PREDICT 'quality' = 6
WITH Precision = 0.739 AND Coverage = 0.1014
IF 'alcohol' <= 10.20 AND 'total sulfur dioxide' > 62.00
THEN PREDICT 'quality' = 5
WITH Precision = 0.757 AND Coverage = 0.181
IF 'sulphates' <= 0.55 AND 'alcohol' <= 11.10 AND 'volatile acidity' > 0.39
THEN PREDICT 'quality' = 5
WITH Precision = 0.705 AND Coverage = 0.192
```

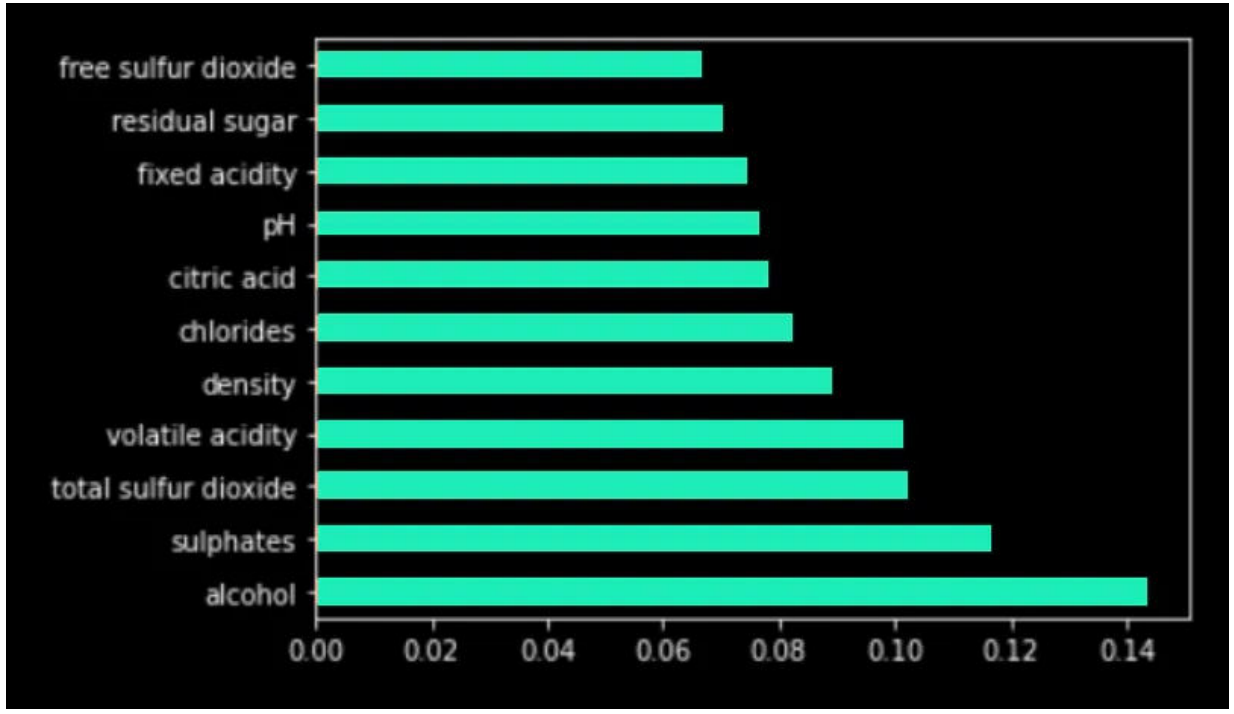
Exploration of models with Anchor method:

To check the relevance of those rules, the variables involved in the rules are compared directly with the results of a feature importance. First, we count the **number of occurrences** of each variable in the rules.



Then, secondly, we run a Feature Importance algorithm on the Random Forest.

```
feature_importances = pd.Series(model.feature_importances_, index=X.columns)
feature_importances.nlargest(15).plot(kind='barh')
```



Rule-Based Anchors (House Prices Dataset)

Concept: Anchors (rule-based explanations)

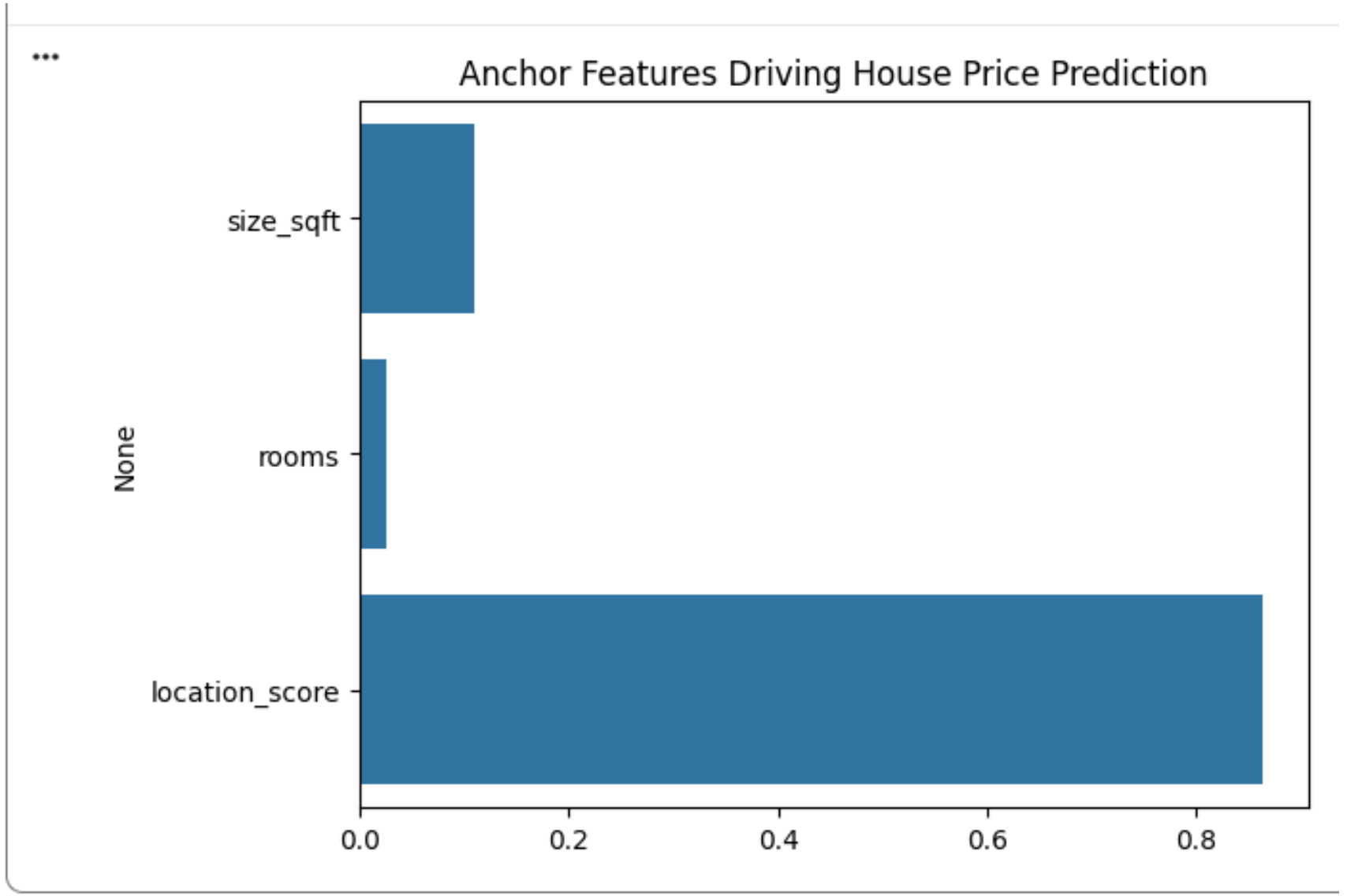
Dataset: house_prices.csv (size, rooms, location_score, price)

Explanation: Anchors are simple rules that strongly influence predictions. Example: "If location_score > 0.8, price is high regardless of other features."

Visualization: Bar chart of feature importance showing anchors like location_score and size_sqft.

Layman's View: Anchors are golden rules: "Big house in good location → expensive."

ANCHORS - Practical



COUNTERFACTUAL EXPLANATIONS (Telecom Churn Dataset)

Concept: Counterfactuals (“what if” scenarios)

Dataset: telecom_churn.csv (tenure, charges, contract_type, churn)

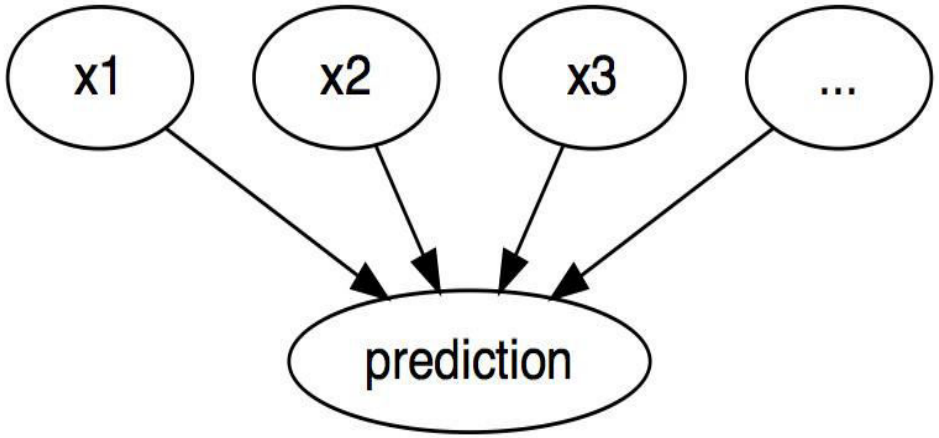
Explanation: Counterfactuals show how changing one feature alters the prediction.
Example: “If contract_type changes from monthly to yearly, churn prediction flips.”

Visualization: Bar chart comparing original vs counterfactual prediction.

Layman’s View: Counterfactuals answer: “What would need to change for a different outcome?”

COUNTER FACTUAL EXPLANATIONS :

- **A counterfactual explanation of a prediction describes the smallest change to the feature values that changes the prediction to a predefined output**.
- A counterfactual explanation of an outcome or a situation takes the form of "**If X had not occurred, Y would not have occurred**".
- Example: "**If I hadn't taken a sip of this hot coffee, I wouldn't have burned my tongue**." Event Y is that I burned my tongue; cause X is that I had a hot coffee.
- Thinking in counterfactuals requires imagining a hypothetical reality that contradicts the observed facts (for example, a world in which I've not drunk the hot coffee), hence the name "counterfactual." The ability to think in counterfactuals makes us humans so smart
- compared to other animals..



- **In interpretable machine learning, counterfactual explanations can be used to explain predictions of individual instances.**
- **The relationship between the inputs and the prediction is very simple: The feature values cause the prediction.**

Counter Factual Explanations:

How It works:

Minimal Perturbation: They find the smallest, most realistic adjustments to an instance's features (e.g., income, age, BMI) that result in a different prediction (e.g., loan approval, non-diabetic).

Contrastive Nature: They contrast the original, rejected outcome with a hypothetical, successful alternative.

Example: If a loan is denied due to low income, a counterfactual might be: "If your income was $5,000 higher, your loan would have been approved"

Key Benefits:

Actionable Insights: Users know what to change to achieve a different result, providing concrete steps for recourse.

Trust & Debugging: Help users trust decisions and allow providers to check for fairness, bias, or model robustness, acting as a debugging tool.

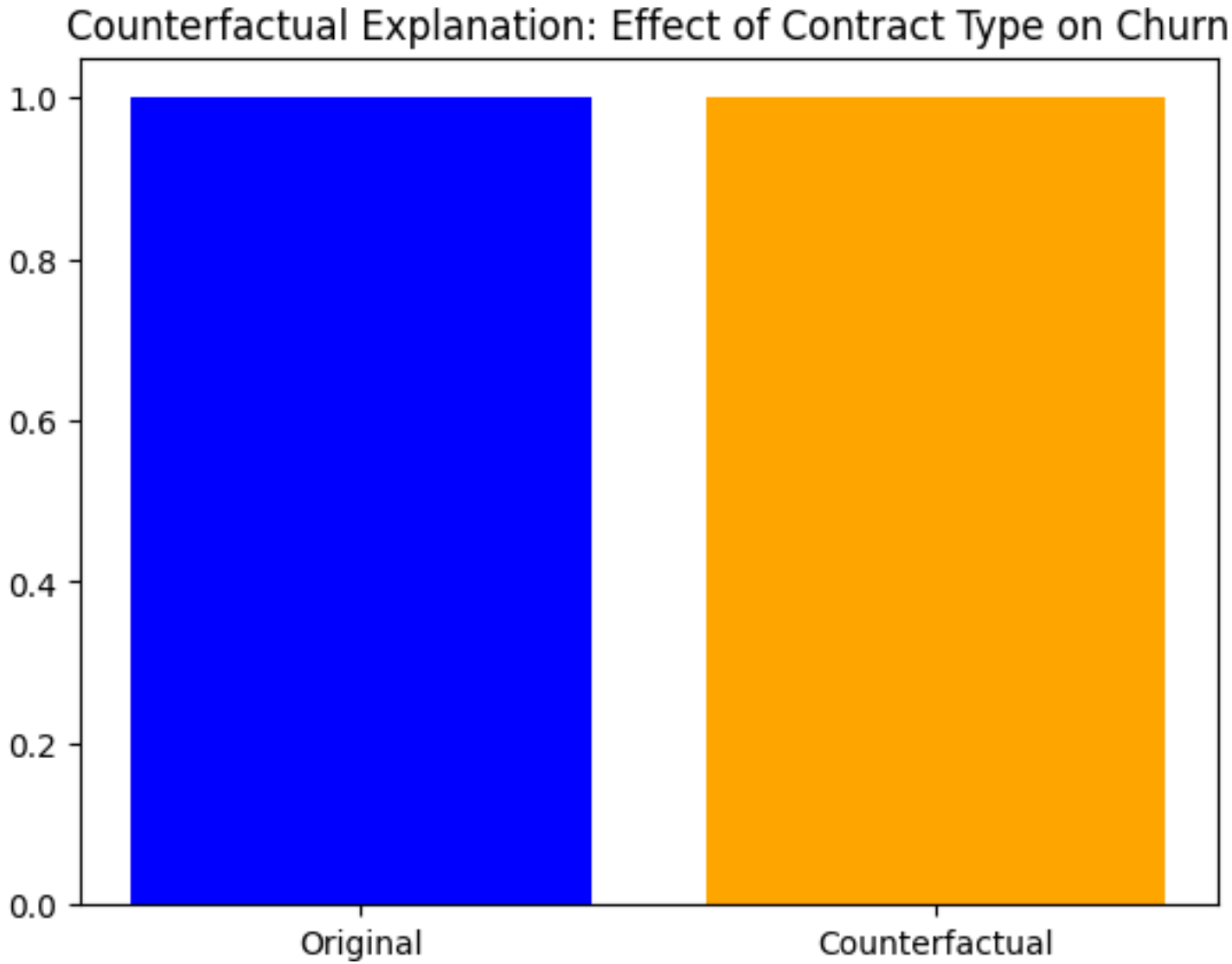
Human-Friendly: They align with how humans reason causally about hypothetical situations, making them easy to understand

✅ **Challenge!** –
Rashomon Effect: Multiple valid, contradictory counterfactuals can exist, requiring methods to select the "best" or most relevant one

Causality vs. Correlation: While intuitive, AI models often learn correlations, not true causation

Counter Factual Explanations – Practical

```
Original prediction: 1
Counterfactual prediction (changed contract_type): 1
```



Counter Factual Explanations – Usecase

- Peter applies for a loan and gets rejected by the (machine learning-powered) banking software. He wonders why his application was rejected and how he might improve his chances to get a loan.
- The question of "why" can be formulated as a counterfactual: What's the smallest change to the features (income, number of credit cards, age, …) that would change the prediction from rejected to approved?
- One possible answer could be: **If Peter would earn 10,000 more per year, he would get the loan**. Or **if Peter had fewer credit cards and had not defaulted on a loan five years ago, he would get the loan**. Peter will never know the reasons for the rejection, as the bank has no interest in transparency, but that is another story.

- 15 Counterfactual Explanations – Interpretable Machine Learning

STRUCTURED DATA EXPLAINERS

Concept: Structured Data Explainers (feature importance)

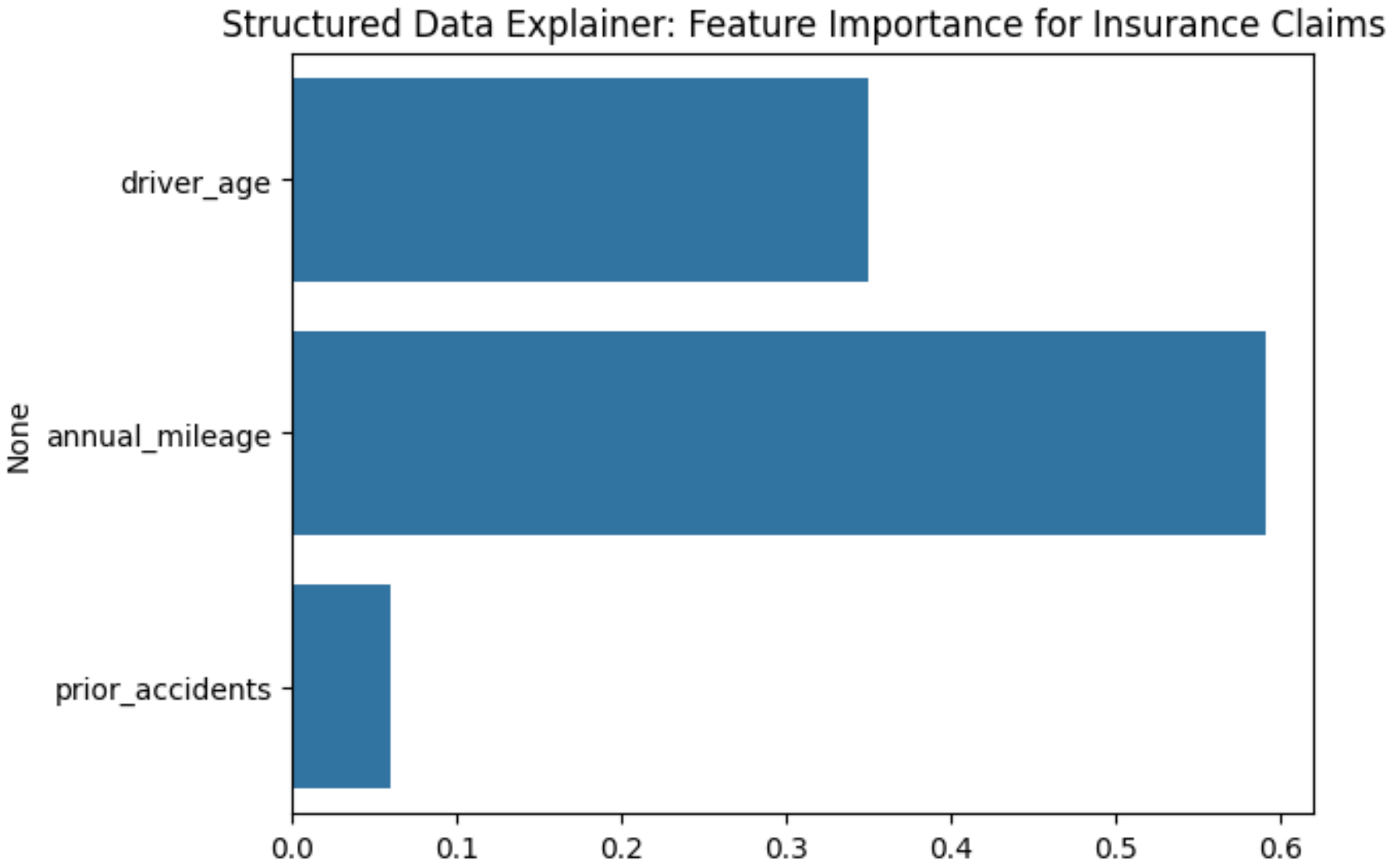
Dataset: insurance_data.csv (driver_age, mileage, prior_accidents, claim)

Explanation: Structured explainers show which features most influence predictions. Example: prior_accidents and mileage are strong predictors of claims.

Visualization: Bar chart of feature importance.

Layman's View: Like ranking risk factors: "More accidents and higher mileage → higher claim probability"

Structured Data Explainers - Practical



Unit 1 – Recap part 1

1. Linear Regression

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n + \epsilon$$

2. Logistic Regression

$$P(y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n)}}$$

3. Generalized Linear Models (GLM)

$$g(\mu) = X\beta$$

4. Generalized Additive Models (GAM

$$y = \beta_0 + f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n) + \epsilon$$

Unit 1 – Recap part 2

5. LIME (Local Interpretable Model-Agnostic Explanations) $\text{argmin}_g \sum_{z \in Z} \pi_x(z) \cdot (f(z) - g(z))^2$

6. Counterfactual Explanations $x^{'} = \text{argmin}_{x'} \, d(x, x^{'}) \quad \text{s.t.} \quad f(x^{'}) \neq f(x)$

7. Decision Tree Splits $\text{IF } x_i \leq t \quad \text{THEN go left, ELSE go right}$

Linear Regression

Equation: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n + \varepsilon$

Coefficients: Each β_i shows how much the target changes when feature x_i increases by 1 unit.

Intercept (β_0): Baseline prediction when all features are zero.

XAI Justification: Intrinsically interpretable — coefficients directly explain predictions.

Example: Each extra square foot adds ₹245 to house price.

Logistic Regression

Equation: $P(y=1|X) = 1 / (1 + e^{-(\beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n)})$

Coefficients: Represent change in log-odds of outcome per unit increase in x_i.

Sigmoid Function: Maps linear combination into probability between 0 and 1.

XAI Justification: Coefficients can be interpreted as odds ratios.

Example: Each year of age increases disease odds by 5%.

Generalized Linear Models (GLM)

Equation: $g(\mu) = X\beta$

Link Function (g): Connects predictors to expected value μ.

Coefficients: Meaning depends on chosen link (logit, log, etc.).

XAI Justification: Extends interpretability to diverse data types while keeping transparency.

Example: Poisson regression with log link: each extra accident increases expected claims multiplicatively.

Generalized Additive Models (GAM)

Equation: $y = \beta_0 + f_1(x_1) + f_2(x_2) + \ldots + f_n(x_n) + \varepsilon$

Functions (f_i): Smooth curves showing how each feature contributes.

Intercept (β_0): Baseline prediction.

XAI Justification: Allows non-linear relationships but remains interpretable by visualizing each feature's effect.

Example: BMI curve: risk rises steeply until BMI=30, then levels off.

Counterfactual Explanations

Equation: $x' = \text{argmin}_{\{x'\}} d(x, x')$ subject to $f(x') \neq f(x)$

Original instance (x): Current input.

Counterfactual (x′): Closest alternative input that changes prediction.

Distance function (d): Ensures minimal change.

XAI Justification: Provides actionable insights — “what needs to change for a different outcome.”

Example: Loan denied at GPA=2.8 → Counterfactual says GPA=3.2 flips to approval.

Decision Tree Splits

Equation/Rule: IF $x_i \leq t$ THEN go left ELSE go right

Threshold (t): Value at which feature splits.

Path Rules: Sequence of splits leads to prediction.

XAI Justification: Fully transparent — every prediction can be traced through rules.

Example: If age > 50 and BP > 130 → predict disease.

THANK YOU

Counterfactual Explanations

(15 Counterfactual Explanations – Interpretable Machine Learning)

Authors: Susanne Dandl & Christoph Molnar

A counterfactual explanation describes a causal situation in the form: "If X had not occurred, Y would not have occurred." For example: "If I hadn't taken a sip of this hot coffee, I wouldn't have burned my tongue." Event Y is that I burned my tongue; cause X is that I had a hot coffee. Thinking in counterfactuals requires imagining a hypothetical reality that contradicts the observed facts (for example, a world in which I've not drunk the hot coffee), hence the name "counterfactual." The ability to think in counterfactuals makes us humans so smart compared to other animals.

In interpretable machine learning, counterfactual explanations can be used to explain predictions of individual instances.[1] Displayed as a graph (Figure 15.1), the relationship between the inputs and the prediction is very simple: The feature values cause the prediction. Even if in reality the relationship between the inputs and the outcome to be predicted might not be causal, we can see the inputs of a model as the cause of the prediction.

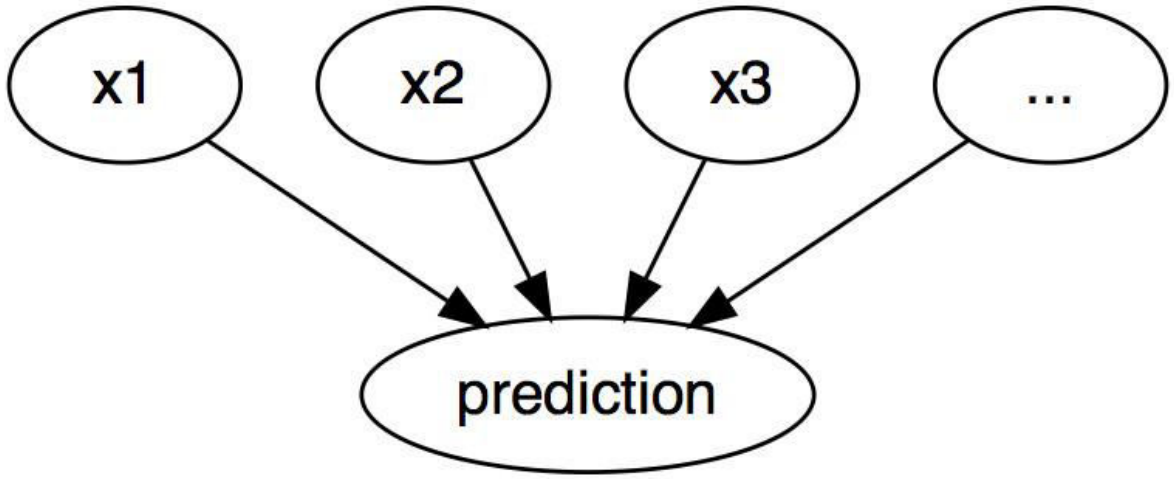


Figure 15.1: The causal relationships between inputs of a machine learning model and the predictions, when the model is merely seen as a black box. The inputs cause the prediction (not necessarily reflecting the real causal relation of the data).

Given this simple graph, it's easy to see how we can simulate counterfactuals for predictions of machine learning models: We simply change the feature values of an instance before making the predictions and we analyze how the prediction changes. We're interested in scenarios in which the prediction changes in a relevant way, like a flip in predicted class (for example, credit application accepted or rejected), or in which the prediction reaches a certain threshold (for example, the probability for cancer reaches 10%). A counterfactual explanation of a prediction describes the smallest change to the feature values that changes the prediction to a predefined output.

There are both model-agnostic and model-specific counterfactual explanation methods, but in this chapter we focus on model-agnostic methods that only work with the model inputs and outputs (and not the internal structure of specific models).

Before discussing how to create counterfactuals, Consider some use cases for counterfactuals and how a good counterfactual explanation looks like.

In this first example, Peter applies for a loan and gets rejected by the (machine learning-powered) banking software. He wonders why his application was rejected and how he might improve his chances to get a loan. The question of "why" can be formulated as a counterfactual: What's the

smallest change to the features (income, number of credit cards, age, …) that would change the prediction from rejected to approved? One possible answer could be: If Peter would earn 10,000 more per year, he would get the loan. Or if Peter had fewer credit cards and had not defaulted on a loan five years ago, he would get the loan. Peter will never know the reasons for the rejection, as the bank has no interest in transparency, but that is another story.

In our second example, we want to explain a model that predicts a continuous outcome with counterfactual explanations. Anna wants to rent out her apartment, but she is not sure how much to charge for it, so she decides to train a machine learning model to predict the rent. Of course, since Anna is a data scientist, that is how she solves her problems. After entering all the details about size, location, whether pets are allowed, and so on, the model tells her that she can charge 900 EUR. She expected 1000 EUR or more, but she trusts her model and decides to play with the feature values of the apartment to see how she can improve the value of the apartment. She finds out that the apartment could be rented out for over 1000 EUR if it were 15 m^2 larger. Interesting, but non-actionable knowledge, because she cannot enlarge her apartment. Finally, by tweaking only the feature values under her control (built-in kitchen yes/no, pets allowed yes/no, type of floor, etc.), she finds out that if she allows pets and installs windows with better insulation, she can charge 1000 EUR. Anna has intuitively worked with counterfactuals to change the outcome. Note that Anna worked with the rent prediction model and wasn't necessarily interested in whether these factors are truly causal for higher rent in the "real world."

Warning

By default, counterfactuals (the IML method) alone don't support causal claims about the real world. This would require a causal model.

Counterfactuals are human-friendly explanations because they are contrastive to the current instance and because they are selective, meaning they usually focus on a small number of feature changes. But counterfactuals suffer from the 'Rashomon effect.' Rashomon is a Japanese movie in which the murder of a Samurai is told by different people. Each of the stories explains the outcome equally well, but the stories contradict each other. The same can also happen with counterfactuals, since there are usually multiple different counterfactual explanations. Each counterfactual tells a different "story" of how a certain outcome was reached. One counterfactual might say to change feature A, the other counterfactual might say to leave A the same but change feature B, which is a contradiction. This issue of multiple truths can be addressed either by reporting all counterfactual explanations or by having a criterion to evaluate counterfactuals and select the best one.

Speaking of criteria, how do we define a good counterfactual explanation? First, the user of a counterfactual explanation defines a relevant change in the prediction of an instance (the alternative reality). An obvious first requirement is that a counterfactual instance produces the predefined prediction as closely as possible. It's not always possible to find a counterfactual with the predefined prediction. For example, in a classification setting with two classes, a rare class and a frequent class, the model might always classify an instance as the frequent class. Changing the feature values so that the predicted label would flip from the frequent class to the rare class might be impossible. We therefore want to relax the requirement that the prediction of the counterfactual must match the predefined outcome exactly. In the classification example, we could look for a counterfactual where the predicted probability of the rare class is increased to 10% instead of the current 2%. The question then is, what are the minimal changes in the features so that the predicted probability changes from 2% to 10% (or close to 10%)?

Tip : Use probabilities

For classification tasks, it's better to define the counterfactual in terms of predicted probabilities than class outcomes.

Another quality criterion is that a counterfactual should be as similar as possible to the instance regarding feature values. The distance between two instances can be measured, for example, with the Manhattan distance or the Gower distance if we have both discrete and continuous features. The counterfactual should not only be close to the original instance, but should also change as few features as possible. To measure how good a counterfactual explanation is in this metric, we can simply count the number of changed features or, in fancy mathematical terms, measure the norm between the counterfactual and actual instance.

Third, it is often desirable to generate multiple diverse counterfactual explanations so that the decision subject gets access to multiple viable ways of generating a different outcome. For instance, continuing our loan example, one counterfactual explanation might suggest only doubling the income to get a loan, while another counterfactual might suggest shifting to a nearby city and increasing the income by a small amount to get a loan. It could be noted that while the first counterfactual might be possible for some, the latter might be more actionable for others. Thus, besides providing a decision subject with different ways to get the desired outcome, diversity also enables "diverse" individuals to alter the features that are convenient for them.

The last requirement is that a counterfactual instance should have feature values that are likely. It would not make sense to generate a counterfactual explanation for the rent example where the size of an apartment is negative or the number of rooms is set to 200. It's even better when the counterfactual is likely according to the joint distribution of the data; for example, an apartment with 10 rooms and 20 m^2 should not be regarded as a counterfactual explanation. Ideally, if the number of square meters is increased, an increase in the number of rooms should also be proposed.

Generating counterfactual explanations

A simple and naive approach to generating counterfactual explanations is searching by trial and error. This approach involves randomly changing feature values of the instance of interest and stopping when the desired output is predicted. Like the example where Anna tried to find a version of her apartment for which she could charge more rent. But there are better approaches than trial and error. First, we define a loss function based on the criteria mentioned above. This loss takes as input the instance of interest, a counterfactual, and the desired (counterfactual) outcome. Then, we can find the counterfactual explanation that minimizes this loss using an optimization algorithm.

Scoped Rules (Anchors)

(16 Scoped Rules (Anchors) – Interpretable Machine Learning)

Authors: Tobias Goerke & Magdalena Lang (with later edits from Christoph Molnar)

The anchors method explains individual predictions of any black box classification model by finding a decision rule that "anchors" the prediction sufficiently. A rule anchors a prediction if changes in other feature values do not affect the prediction. Anchors utilizes reinforcement learning techniques in

combination with a graph search algorithm to reduce the number of model calls (and hence the required runtime) to a minimum while still being able to recover from local optima. Ribeiro, Singh, and Guestrin (2018) proposed the anchors algorithm – the same researchers who introduced the LIME algorithm.

Like its predecessor, the anchors approach deploys a *perturbation-based* strategy to generate *local* explanations for predictions of black box machine learning models. However, instead of surrogate models used by LIME, the resulting explanations are expressed as easy-to-understand *IF-THEN* rules, called *anchors*. These rules are reusable since they are *scoped*: anchors include the notion of coverage, stating precisely to which other, possibly unseen, instances they apply. Finding anchors involves an exploration or multi-armed bandit problem, which originates in the discipline of reinforcement learning. To this end, neighbors, or perturbations, are created and evaluated for every instance that is being explained. Doing so allows the approach to disregard the black box's structure and its internal parameters so that these can remain both unobserved and unaltered. Thus, the algorithm is *model-agnostic*, meaning it can be applied to **any** class of model.

In their paper, the authors compare both of their algorithms and visualize how differently these consult an instance's neighborhood to derive results. For this, Figure 16.1 depicts both LIME and anchors locally explaining a complex binary classifier (predicts either or) using two exemplary instances. LIME's results do not indicate how faithful they are, as LIME solely learns a linear decision boundary that best approximates the model given a perturbation space . Given the same perturbation space, the anchors approach constructs explanations whose coverage is adapted to the model's behavior, and the approach clearly expresses their boundaries. Thus, they are faithful by design and state exactly for which instances they are valid. This property makes anchors particularly intuitive and easy to comprehend.

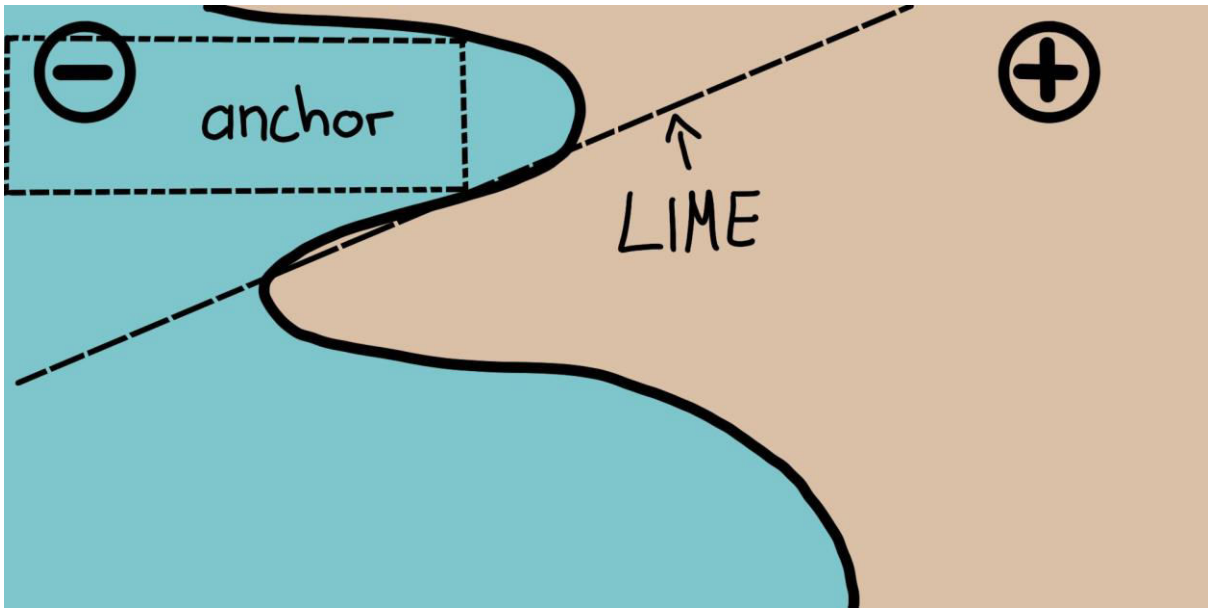


Figure 16.1: LIME vs. Anchors – A Toy Visualization. Inspired by figure from Ribeiro, Singh, and Guestrin (2018).

As mentioned before, the algorithm's results or explanations come in the form of rules, called anchors. The following simple example illustrates such an anchor. For instance, suppose we are given a bivariate black box model that predicts whether or not a passenger survived the Titanic disaster. Now we would like to know *why* the model predicts for one specific person (see Table 16.1) that they survived. The anchors algorithm provides a result explanation like the one shown below.

Table 16.1: Example to be explained

Feature	Value
Age	20
Sex	female
Class	first
Ticket price	300$
More attributes	…
Survived	true

And the corresponding anchors explanation is:

IF SEX = female AND Class = first THEN PREDICT Survived = true WITH PRECISION 97% AND COVERAGE 15%

The example shows how anchors can provide essential insights into a model's prediction and its underlying reasoning. The result shows which attributes were taken into account by the model, which in this case, are female and first class. Humans, being paramount for correctness, can use this rule to validate the model's behavior. The anchor additionally tells us that it applies to 15% of the perturbation space's instances. In those cases, the explanation is 97% accurate, meaning the displayed predicates are almost exclusively responsible for the predicted outcome.

An anchor is formally defined as follows:

$$\mathbb{E}_{\mathcal{D}_{\mathbf{x}}(\mathbf{z}|A)}[1_{\hat{f}(\mathbf{x})=\hat{f}(\mathbf{z})}] \geq \tau; A(\mathbf{x}) = 1$$

Wherein:

- represents the instance being explained (e.g. one row in a tabular dataset).
- is a set of predicates, i.e., the resulting rule or anchor, such that when all feature predicates defined by correspond to 's feature values.
- denotes the classification model to be explained (e.g. an artificial neural network model). It can be queried to predict a label for and its perturbations.
- indicates the distribution of neighbors of , matching .
- specifies a precision threshold. Only rules that achieve a local fidelity of at least are considered a valid result.

The formal description may be intimidating and can be put in words:

Given an instance $\mathbf{x}$ to be explained, a rule or an anchor A is to be found, such that it applies to $\mathbf{x}$, while the same class as for $\mathbf{x}$ gets predicted for a fraction of at least τ of $\mathbf{x}$'s neighbors where the same A is applicable. A rule's precision results from evaluating neighbors or perturbations (following $\mathcal{D}_{\mathbf{x}}(\mathbf{z}|A)$) using the provided machine learning model (denoted by the indicator function $1_{\hat{f}(\mathbf{x})=\hat{f}(\mathbf{z})}$).

Tip :Carefully set precision

Set the precision threshold T wisely: A higher ensures stronger rules, but may reduce coverage. Experiment with different values to balance precision and coverage.

Finding anchors

Although anchors' mathematical description may seem clear and straightforward, constructing particular rules is infeasible. It would require evaluating $1_{\hat{f}(\mathbf{x})=\hat{f}(\mathbf{z})} \text{ for all } \mathbf{z} \in \mathcal{D}_{\mathbf{x}}(\cdot|A),$ for all , which is not possible in continuous or large input spaces. Therefore, the authors propose to introduce the parameter $0 \leq \delta \leq 1$ to create a probabilistic definition. This way, samples are drawn until there is statistical confidence concerning their precision. The probabilistic definition reads as follows:

The previous two definitions are combined and extended by the notion of coverage. Its rationale consists of finding rules that apply to a preferably large part of the model's input space. Coverage is formally defined as an anchor's probability of applying to its neighbors, i.e., its perturbation space:

Including this element leads to the anchor's final definition taking into account the maximization of coverage:

Thus, the proceeding strives for a rule that has the highest coverage among all eligible rules (all those that satisfy the precision threshold given the probabilistic definition). These rules are thought to be more important, as they describe a larger part of the model. Note that rules with more predicates tend to have higher precision than rules with fewer predicates. In particular, a rule that fixes every feature of reduces the evaluated neighborhood to identical instances. Thus, the model classifies all neighbors equally, and the rule's precision is . At the same time, a rule that fixes many features is overly specific and only applicable to a few instances. Hence, there is a *trade-off between precision and coverage*.

The anchors approach uses four main components to find explanations.

Candidate Generation: Generates new explanation candidates. In the first round, one candidate per feature of gets created and fixes the respective value of possible perturbations. In every other round, the best candidates of the previous round are extended by one feature predicate that is not yet contained therein.

Best Candidate Identification: Candidate rules are to be compared in regard to which rule explains the best. To this end, perturbations that match the currently observed rule are created and evaluated by calling the model. However, these calls need to be minimized to limit computational overhead. This is why, at the core of this component, there is a pure-exploration Multi-Armed Bandit (*MAB*). To be more precise, it's *KL-LUCB* by Kaufmann and Kalyanakrishnan (2013). MABs are used to efficiently explore and exploit different strategies (called arms in an analogy to slot machines) using sequential selection. In the given setting, each candidate rule is to be seen as an arm that can be pulled. Each time it is pulled, respective neighbors get evaluated, and we thereby obtain more information about the candidate rule's payoff (precision in the anchor's case). The precision thus states how well the rule describes the instance to be explained.

Candidate Precision Validation: Takes more samples in case there is no statistical confidence yet that the candidate exceeds the threshold.

Modified Beam Search: All of the above components are assembled in a beam search, which is a graph search algorithm and a variant of the breadth-first algorithm. It carries the best candidates of each round over to the next one (where is called the *Beam Width*). These best rules are then used to create new rules. The beam search conducts at most rounds, as each feature can only be included in a rule at most once. Thus, at every round , it generates candidates with exactly predicates and selects the best thereof. Therefore, by setting high, the algorithm is more likely to avoid local optima. In turn, this requires a high number of model calls and thereby increases the computational load.

These four components are shown in Figure 16.2.

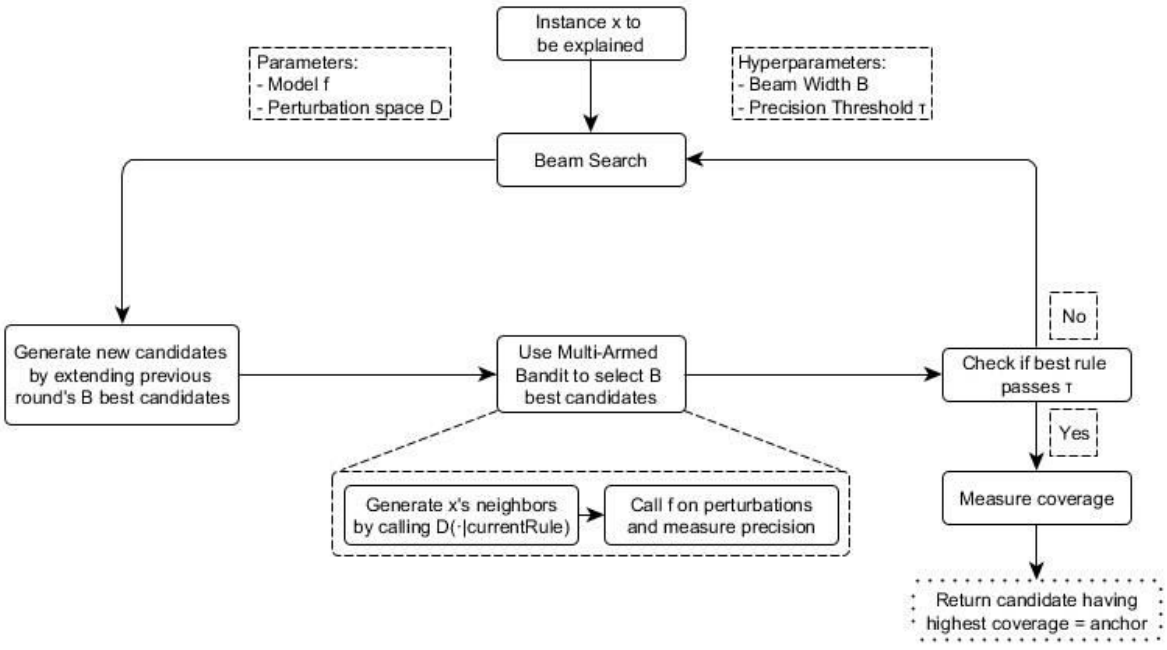


Figure 16.2: The anchors algorithm's components and their interrelations (simplified)

The approach is a seemingly perfect recipe for efficiently deriving statistically sound information about why any system classified an instance the way it did. It systematically experiments with the model's input and concludes by observing respective outputs. It relies on well-established and researched Machine Learning methods (MABs) to reduce the number of calls made to the model. This, in turn, drastically reduces the algorithm's runtime.

Complexity and runtime

Knowing the anchors approach's asymptotic runtime behavior helps to evaluate how well it is expected to perform on specific problems. Let denote the beam width and the number of all features. Then the anchors algorithm is subject to:

This boundary abstracts from problem-independent hyperparameters, such as the statistical confidence . Ignoring hyperparameters helps reduce the boundary's complexity (see original paper for more info). Since the MAB extracts the best out of candidates in each round, most MABs and their runtimes multiply the factor more than any other parameter.

It thus becomes apparent: the algorithm's efficiency decreases when many features are present.

Tabular data example

Tabular data is structured data represented by tables, wherein columns embody features and rows instances. For instance, we use the bike rental data to demonstrate the anchors approach's potential to explain machine learning predictions for selected instances. For this, we turn the regression into a classification problem and train a random forest as our black box model. It's to classify whether the number of rented bikes lies above or below the trend line. We also use additional features like the number of days since 2011 as a time trend, the month, and the weekday.

Before creating anchor explanations, one needs to define a perturbation function. An easy way to do so is to use an intuitive default perturbation space for tabular explanation cases, which can be built by sampling from, e.g., the training data. When perturbing an instance, this default approach maintains the features' values that are subject to the anchors' predicates, while replacing the non-fixed features with values taken from another randomly sampled instance with a specified probability. This process yields new instances that are similar to the explained one but have adopted some values from other random instances. Thus, they resemble neighbors of the explained instance.

The results (Figure 16.3) are instinctively interpretable and show, for each explained instance, which features are most important for the model's prediction. As the anchors only have a few predicates, they additionally have high coverage and hence apply to other cases. The rules shown above were generated with . Thus, we ask for anchors whose evaluated perturbations faithfully support the label with an accuracy of at least . Also, discretization was used to increase the expressiveness and applicability of numerical features.

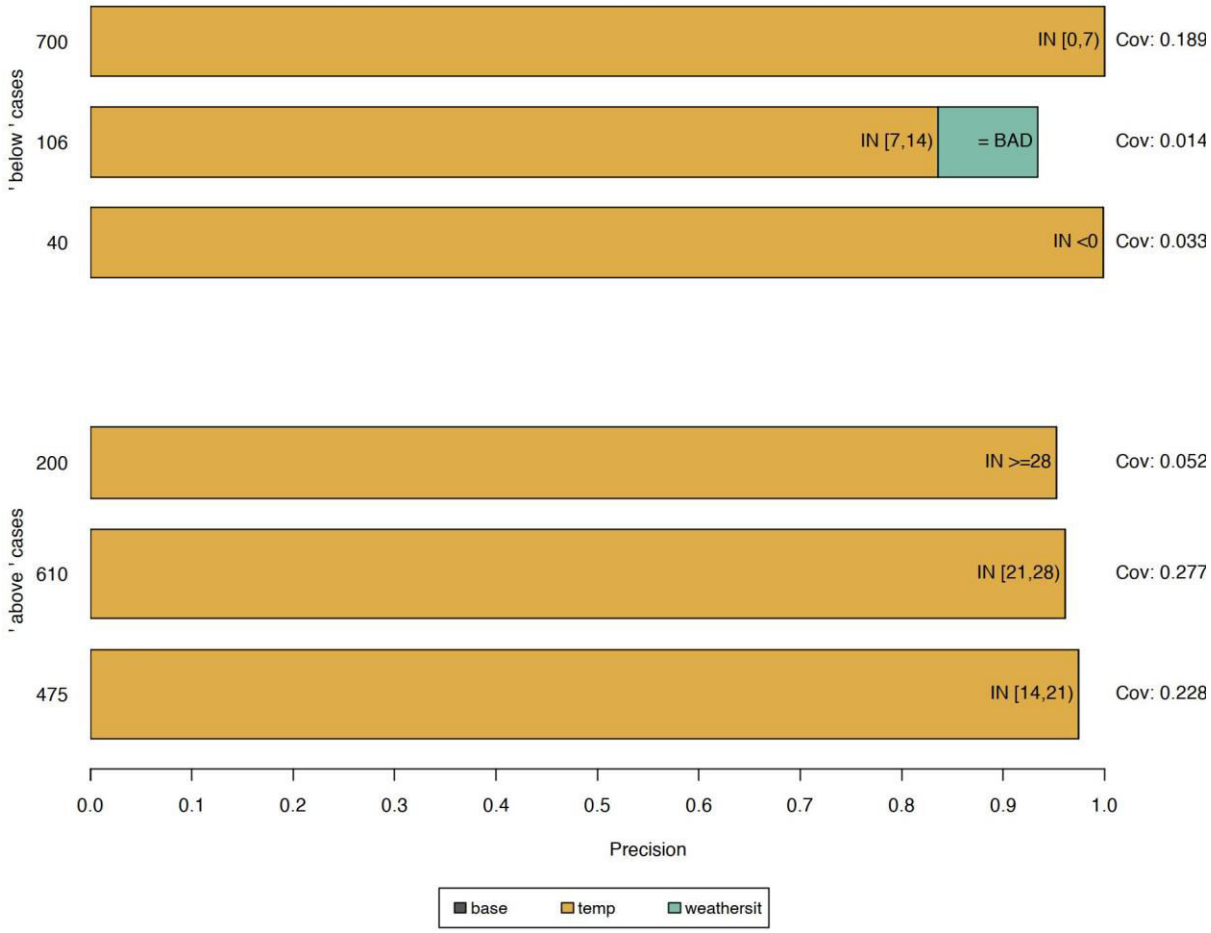


Figure 16.3: Anchors explaining six instances of the bike rental dataset. Each row represents one explanation or anchor, and each bar depicts the feature predicates contained by it. The x-axis displays a rule's precision, and a bar's thickness corresponds to its coverage. The "base" rule contains no predicates. These anchors show that the model mainly considers the temperature for predictions.

All of the previous rules were generated for instances where the model decides confidently based on a few features. However, other instances are not as distinctly classified by the model, as more features are of importance. In such cases, anchors get more specific, comprise more features, and apply to fewer instances, as for example for Figure 16.4.

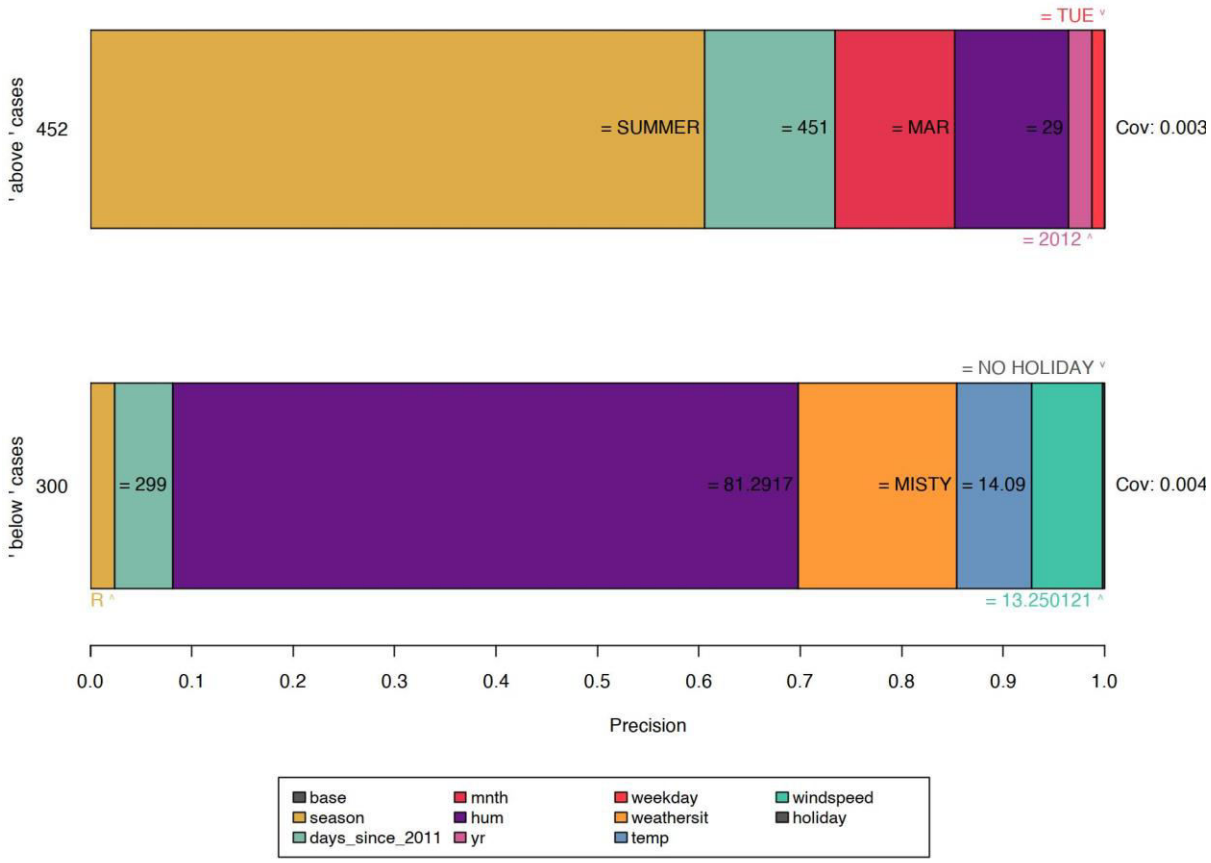


Figure 16.4: Explaining instances near decision boundaries leads to specific rules comprising a higher number of feature predicates and lower coverage. Also, the empty rule, i.e., the base feature, gets less important. This can be interpreted as a signal for a decision boundary, as the instance is located in a volatile neighborhood.

While choosing the default perturbation space is a comfortable choice to make, it may have a great impact on the algorithm and can thus lead to biased results. For example, if the training set is unbalanced (there is an unequal number of instances of each class), the perturbation space is as well. This condition further affects the rule-finding and the result's precision. This may be unwanted and can be approached in multiple ways.

For example, a custom perturbation space can be defined. This custom perturbation can sample differently, e.g., from an unbalanced data set or a normal distribution. This, however, comes with a side effect: the sampled neighbors are not representative and change the coverage's scope. Alternatively, we could modify the MAB's confidence and error parameter values . This would cause the MAB to draw more samples, ultimately leading to the minority being sampled more often in absolute terms.

TipCheck for class imbalance

When defining a custom perturbation space, consider sampling strategies that respect the class imbalance. For example, use stratified sampling to ensure that minority classes are appropriately represented in the perturbation space.

Strengths

The anchors approach offers multiple advantages over LIME. First, the algorithm's output is easier to understand, as rules are **easy to interpret** (even for laypersons).

Furthermore, **anchors are subsettable** and even state a measure of importance by including the notion of coverage. Second, the anchors approach **works when model predictions are non-linear or complex** in an instance's neighborhood. As the approach deploys reinforcement learning techniques instead of fitting surrogate models, it is less likely to underfit the model.

Apart from that, the algorithm is **model-agnostic** and thus applicable to any model.

Anchors can be useful for justifying predictions as they **show the robustness of predictions** (or lack thereof) to changes in certain features.

Furthermore, it is **highly efficient as it can be parallelized** by making use of MABs that support batch sampling (e.g. BatchSAR).

Limitations

The algorithm suffers from a **highly configurable** and impactful setup, just like most perturbation-based explainers. Not only do hyperparameters such as the beam width or precision threshold need to be tuned to yield meaningful results, but also the perturbation function needs to be explicitly designed for one domain/use case. Think of how tabular data get perturbed, and think of how to apply the same concepts to image data (hint: these cannot be applied). Luckily, default approaches may be used in some domains (e.g., tabular), facilitating an initial explanation setup.

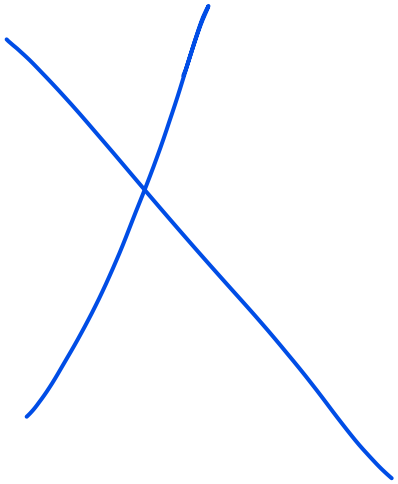
Also, **many scenarios require discretization** as otherwise results are too specific, have low coverage, and do not contribute to understanding the model. While discretization can help, it may also blur decision boundaries if used carelessly and thus have the exact opposite effect. Since there is no best discretization technique, users need to be aware of the data before deciding on how to discretize data, not to obtain poor results.

Constructing anchors requires **many calls to the machine learning model**, just like all perturbation-based explainers. While the algorithm deploys MABs to minimize the number of calls, its runtime still very much depends on the model's performance and is therefore highly variable.

Lastly, the notion of **coverage is undefined in some domains**. For example, there is no obvious or universal definition of how superpixels in one image compare to those in other images.

Software and alternatives

Currently, there are two implementations available: anchor, a Python package (also integrated by Alibi), and a Java implementation. The former is the anchors algorithm's authors' reference, and the latter a high-performance implementation which comes with an R interface, called anchors, which was used for the examples in this chapter. By now, the anchors implementation supports tabular data only. However, anchors may theoretically be constructed for any domain or type of data.



Fundamentals of Generalized Additive Models

Definition of GAMs

Generalized Additive Models (GAMs) are a versatile statistical modelling technique used to analyze complex relationships within data. Unlike linear models, GAMs can capture non-linear patterns by combining multiple smooth functions of predictor variables. Generalized Additive Models (GAM Models) are particularly valuable when investigating intricate dependencies, making them a crucial tool for data analysis and **predictive modeling**.

Differences between GAMs and Linear Regression

Aspect	**Generalized Additive Models (GAMs)**	**Linear Regression**
Modeling Assumption	Flexible; no assumption of linearity between predictors and the response variable.	Assumes a linear relationship between predictors and the response variable.
Model Flexibility	Can capture complex, non-linear relationships between predictors and the response.	Limited to modeling linear relationships; may not handle non-linearity well.
Parametric vs. Non-Parametric	Non-parametric: does not require a predefined functional form.	Parametric: assumes a specific functional form (e.g., linear).
Model Complexity	Can be highly complex, accommodating intricate relationships.	Simpler in terms of model structure due to linearity assumption.
Interpretability	Provides interpretable results, especially when examining smooth functions.	Interpretation is straightforward but may lack detail for complex relationships.
Regularization	Can include regularization techniques to control model complexity.	ularization methods like Ridge or
Data Handling	Tolerant of missing data and can handle it effectively.	Missing data handling is less straightforward; imputation may be necessary.
Sample Size Requirements	May require larger sample sizes to capture non-linear patterns effectively.	Less stringent sample size requirements due to simpler model assumptions.

Model Complexity Management	Manages complexity through the choice of smoothing functions and regularization.	Complexity management relies on feature selection and external techniques.
Assumption Testing	Assumes fewer assumptions about the data distribution, making it more robust.	Assumes specific distributional properties, which can lead to violations.
Visualizations	Visualization of smooth functions aids in interpreting relationships.	Visualizations are limited to scatterplots and linear trends.
Applications	Versatile and suitable for various data types, including both regression and classification tasks.	Primarily used for linear regression tasks; extensions required for classification.

Advantages and Disadvantages of Generalized Additive Models (GAModels)

Sr. No.	Advantages of GAMs	Disadvantages of GAMs
1.	**Flexibility:** GAMs can model various relationships, including non-linear and complex patterns.	**Complexity:** GAMs can become computationally intensive for large datasets or high-dimensional problems.
2.	**Interpretability:** They provide interpretable results, making understanding the relationships between predictors and the response easier.	**Data Requirements:** GAMs may require larger sample sizes to capture non-linear patterns effectively.
3.	**Non-linearity:** GAMs can capture intricate, non-linear relationships that traditional linear models cannot represent.	**Sensitivity to Smoothing Parameters:** The choice of smoothing parameters can impact model results, requiring careful tuning.
4.	**Regularization:** GAMs can incorporate regularization techniques to prevent overfitting and improve generalization.	**Model Selection:** Selecting the appropriate number and type of smooth terms can be challenging.
5.	**Visualization:** The smooth functions in GAMs can be visually represented, aiding in model interpretation.	**Limited to Regression and Classification:** GAMs are primarily suited for regression and classification tasks and may not be suitable for more complex tasks like image recognition.

Building Generalized Additive Models

Building Generalized Additive Models (GAMs) is a multi-step process that involves **data preparation**, variable selection, fitting the model, and validating its

performance. Here, we'll delve into these essential steps to guide you in constructing accurate and reliable GAMs.

Data Preparation for GAM Models

- **Handling Missing Data:** Address any missing values in your dataset. GAMs can **accommodate missing data** points, but handling them appropriately through imputation or modeling strategies is essential.
- **Encoding Categorical Variables:** If your dataset includes categorical predictors, encode them into a numeric format using techniques like one-hot encoding or label encoding.
- **Scaling Numeric Features:** Standardize or scale numeric features to ensure the model treats them fairly. Common scaling methods include z-score standardization or min-max scaling.

Selecting Appropriate Variables and Features

- **Domain Knowledge:** Start by considering your domain knowledge. Which predictors are likely to influence the response variable? This qualitative understanding can guide your variable selection process.
- **Feature Engineering:** Create new features that might capture important relationships or interactions. For instance, you can generate polynomial features or interaction terms between variables.
- **Feature Selection:** Use techniques like feature importance, recursive feature elimination, or regularization (e.g., **Lasso**) to identify the most relevant predictors. Reducing the dimensionality of your feature space can improve model simplicity and generalization.

Techniques for Fitting and Validating GAMs

Choosing Smoothing Functions: Generalized Additive Models use smoothing functions to model relationships between predictors and the response. Based on the nature of your data and the expected relationships, select appropriate smoothing functions, such as cubic splines or thin-plate splines.

Cross-Validation: Employ techniques like k-fold cross-validation to assess your model's generalization performance. This helps in detecting overfitting and guides hyperparameter tuning.

Regularization: Apply regularization techniques, like penalty terms (e.g., ridge or Lasso), to control the complexity of the GAM and prevent **overfitting**. These techniques can help balance fitting the data well and avoiding excessive complexity.

Model Selection: Experiment with different model configurations, including the number and type of smooth terms. Model selection criteria such as AIC or BIC can assist in choosing the optimal model.

Best Practices for Building Accurate and Reliable GAMs

1. **Balance Interpretability and Complexity:** While GAMs are flexible, they strive to balance model complexity and interpretability. Simpler models are often more interpretable and generalize better.
2. **Regularize When Necessary:** Apply regularization when dealing with noisy or high-dimensional data to improve model stability and reduce the risk of overfitting.
3. **Visualize the Data:** Create visualizations of your data and model output. Visualization can help you understand the relationships modeled by the GAM and communicate insights effectively.
4. **Test Assumptions:** Ensure that the assumptions of the Generalized Additive Models (GAM Models), such as the linearity of smooth terms, are met. Diagnostic plots and residual analysis can help identify any violations.

Interpreting Generalized Additive Models

Interpreting Generalized Additive Models (GAMs) is crucial for extracting meaningful insights from the model's output. Here, we'll explore techniques for understanding and communicating GAM results effectively.

Understanding the Output of GAMs

Smooth Functions: GAMs produce smooth functions for each predictor variable, showing how they influence the response variable. These functions are often displayed graphically and represent the estimated relationships.

Estimated Parameters: Examine the estimated coefficients for each smooth term. These coefficients indicate the strength and direction of the relationship between the predictor

and the response. Positive coefficients imply a positive association, while negative coefficients suggest a negative association.

Deviance Explained: GAM Models output a measure of deviance explained by the model. A higher percentage of deviance explained indicates a better fit of the model to the data.

Techniques for Visualizing GAM Results

1. **Partial Dependence Plots (PDPs):** Create PDPs to visualize the effect of one predictor while keeping others constant. PDPs help understand how a predictor influences the response across its range.
2. **Interaction Plots:** Generate interaction plots to explore the interactions between two or more predictors. These plots show how the relationship between predictors and the response changes based on the values of other predictors.
3. **Component-Wise Plots:** Component-wise plots display the contributions of each smooth term to the overall prediction. These plots can highlight which terms have the most significant impact.
4. **Residual Plots:** Examine residual plots to assess the model's goodness of fit. Deviations from randomness in residuals may indicate unaccounted-for patterns or model misspecification.

Techniques for Interpreting GAM Results

1. **Identify Significance:** Determine which smooth terms are statistically significant. Techniques like hypothesis tests or confidence intervals can help assess the significance of terms.

2. **Understanding Shapes:** Focus on the shapes of the smooth functions. Look for inflection points, non-linearities, or unusual patterns. These shapes provide insights into the relationships within the data.
3. **Interaction Interpretation:** When interactions are present, interpret how the relationship between one predictor and the response changes with different values of another predictor.
4. **Quantify Effects:** If applicable, quantify the effects of predictors on the response. For example, you can estimate the change in the response for a one-unit change in a predictor.

Best Practices for Communicating GAM Results to Non-Technical Stakeholders

- **Simplify the Message:** Translate technical terms and jargon into plain language. Focus on conveying the key findings and insights without overwhelming stakeholders with technical details.
- **Use Visual Aids:** Visualizations are powerful tools for communication. Share plots, graphs, and charts that clearly illustrate the model's results.
- **Provide Context:** Place the results in context by explaining the real-world implications of the findings. How do the model's insights impact decision-making or business strategies?
- **Highlight Certainty:** Be transparent about the uncertainties associated with the model's predictions. Communicate confidence intervals or prediction intervals to convey the range of possible outcomes.
- **Address Limitations:** Acknowledge the limitations of the model. Discuss any assumptions made and potential sources of error or bias.

Applications of Generalized Additive Models

Let us explore the applications of Generalized Additive Models (GAMs) across various industries, through use cases and case studies.

Use Cases of GAMs in Different Industries

Generalized Additive Models (GAMs) find application across various industries and domains due to their ability to model complex relationships in data. Here are some key applications:

1. Healthcare:

- Predicting patient outcomes based on medical variables.
- Analyzing the effects of environmental factors on public health.

2. Finance:

- Modeling financial risk and predicting market trends.
- Credit scoring and assessing loan default risks.

3. Environmental Science:

- Studying climate change and its impact on ecosystems.
- Analyzing air and water quality data to identify trends.

4. Marketing:

- Optimizing advertising campaigns by modeling customer response.
- Predicting customer churn and segmenting customer populations.

5. Ecology:

- Modeling species distribution and habitat suitability.

- Studying the impact of environmental factors on biodiversity.

6. Manufacturing:

- Predictive maintenance to reduce equipment downtime.
- Quality control and defect detection in production processes.

7. Social Sciences:

- Analyzing survey data to study social trends and behaviors.
- Assessing the impact of educational interventions on student performance.

Comparison of GAMs with Other Machine Learning Techniques

Generalized Additive Models (GAMs)	**Other Machine Learning Techniques**
Semi-parametric; combines linear and non-linear components.	Varies widely, including decision trees, random forests, support vector machines, neural networks, etc.
Highly interpretable; provides insights into relationships between predictors and the response.	Interpretability varies; some models, like decision trees, are interpretable, while others, like neural networks, are less so.
Well-suited for capturing non-linear relationships between predictors and the response.	Capable of handling non-linearity to varying degrees, depending on the technique.
Can include regularization techniques to control model complexity.	Regularization techniques are often employed in other models (e.g., L1 and L2 regularization in neural networks).
Complexity management through the choice of smoothing functions and regularization.	Complex models may require careful tuning to prevent overfitting.
May require larger sample sizes to capture non-linear patterns effectively.	Data requirements vary by technique but generally depend on the model's complexity.
Generally less computationally intensive than some deep learning methods.	Deep learning models can be computationally intensive, especially for large-scale applications.
Relatively straightforward to implement and understand, making them accessible.	Implementation complexity varies, with some techniques requiring specialized libraries and expertise.
Involves selecting the number and type of smooth terms and tuning smoothing parameters.	Model selection and hyperparameter tuning are integral and vary by technique.

Tolerant of missing data and can handle it effectively.	Handling missing data varies, with some models requiring imputation or other strategies.
Versatile, suitable for various data types, including regression and classification tasks.	Diverse applications, including image recognition (convolutional neural networks), natural language processing (recurrent neural networks), and more.
Scalability depends on the data size and complexity but generally can handle medium-sized datasets well.	Scalability varies by technique, with some models capable of handling large-scale data (e.g., gradient boosting).

Case Studies of Successful Applications of GAMs

Environmental Modeling: GAMs have been used to study the relationship between climate variables and species distribution. For example, Application of a generalized additive model (GAM) to reveal **relationships between environmental factors and distributions of pelagic fish** and krill: a case study in Sendai Bay, Japan.

Healthcare: **Statistical modeling of COVID-19 data**. In the COVID-19 period, Generalized Additive Models (GAMs) have been successfully employed on many occasions to obtain vital data-driven insights.

Future Potential of GAMs in Research and Business

The future of GAM Models holds significant promises:

- **Advanced Interpretability:** Developments in model interpretation techniques will enhance GAMs' ability to provide actionable insights.
- **Automated Smoothing Parameter Tuning:** Automation tools will simplify the process of choosing optimal smoothing parameters, reducing user burden.
- **Integration with Deep Learning:** Combining the flexibility of GAMs with the power of deep learning can lead to more accurate and interpretable models.

- **Real-time Applications:** GAMs will likely play a pivotal role in real-time decision-making applications across industries, including autonomous vehicles and personalized medicine.

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DEMO: Partial Dependence Plot (PDP)

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Context: Explainable AI - UNIT 2 (Global Model-Agnostic Methods)

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This demo shows how PDP and ALE help us understand

how features influence model predictions globally.

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 # STEP 1: Import libraries

```
import pandas as pd

import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split

from sklearn.ensemble import RandomForestRegressor,
RandomForestClassifier

from sklearn.inspection import PartialDependenceDisplay

import numpy as np
```

 # PART A: Partial Dependence Plot (PDP)

```
# Dataset: house_prices.csv

# Goal: Predict house price based on size, rooms, and location_score
```

```
# ----------------------------
# Load dataset
house_df = pd.read_csv("house_prices.csv")
# Separate features (X) and target (y)
X_house = house_df.drop("price", axis=1)
y_house = house_df["price"]
# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X_house, y_house,
random_state=42)
# Train a Random Forest Regressor
rf_reg = RandomForestRegressor(random_state=42)
rf_reg.fit(X_train, y_train)
# Generate PDP for 'size_sqft' feature
# PDP shows the average effect of house size on predicted price
fig, ax = plt.subplots(figsize=(6, 4))
PartialDependenceDisplay.from_estimator(rf_reg, X_train, ['size_sqft'],
# feature to explain
    feature_names=X_house.columns,   ax=ax)
plt.title("Partial Dependence Plot (House Size vs Price)")
plt.show()
# ----------------------------
#  Note:
# PDP is like asking: “On average, how does house size affect price?”
# It shows a smooth curve: bigger houses → higher predicted prices.
# BUT it assumes features are independent, so may be misleading if
# size and rooms are strongly correlated.
# ----------------------------
```

 # DEMO: Accumulated Local Effects (ALE)

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
# Load dataset
df = pd.read_csv("disease_risk.csv")
X = df.drop("disease", axis=1)
y = df["disease"]
# Train model
X_train, X_test, y_train, y_test = train_test_split(X, y, random_state=42)
clf = RandomForestClassifier(random_state=42)
clf.fit(X_train, y_train)
# Manual ALE for one feature (e.g., 'age')
def ale_1d(model, X, feature, bins=10):
    values = X[feature].values
    percentiles = np.percentile(values, np.linspace(0, 100, bins+1))
    effects = []
```

```
    centers = []
    for i in range(len(percentiles)-1):
        low, high = percentiles[i], percentiles[i+1]
        mask = (values >= low) & (values < high)
        if not np.any(mask):
            continue
        X_low, X_high = X.copy(), X.copy()
        X_low.loc[mask, feature] = low
        X_high.loc[mask, feature] = high
        preds_low =
model.predict_proba(X_low)[:,1][mask]
        preds_high =
model.predict_proba(X_high)[:,1][mask]
        effects.append(np.mean(preds_high - preds_low))
        centers.append((low+high)/2)
    # Accumulate local effects
    ale = np.cumsum(effects) -
np.mean(np.cumsum(effects))
    return centers, ale
centers, ale_vals = ale_1d(clf, X_train, "age", bins=10)
plt.plot(centers, ale_vals, marker="o")
plt.xlabel("Age")
```

```
plt.ylabel("ALE (Effect on Disease Risk)")
plt.title("Accumulated Local Effects for Age")
plt.grid(True)
plt.show()
```